

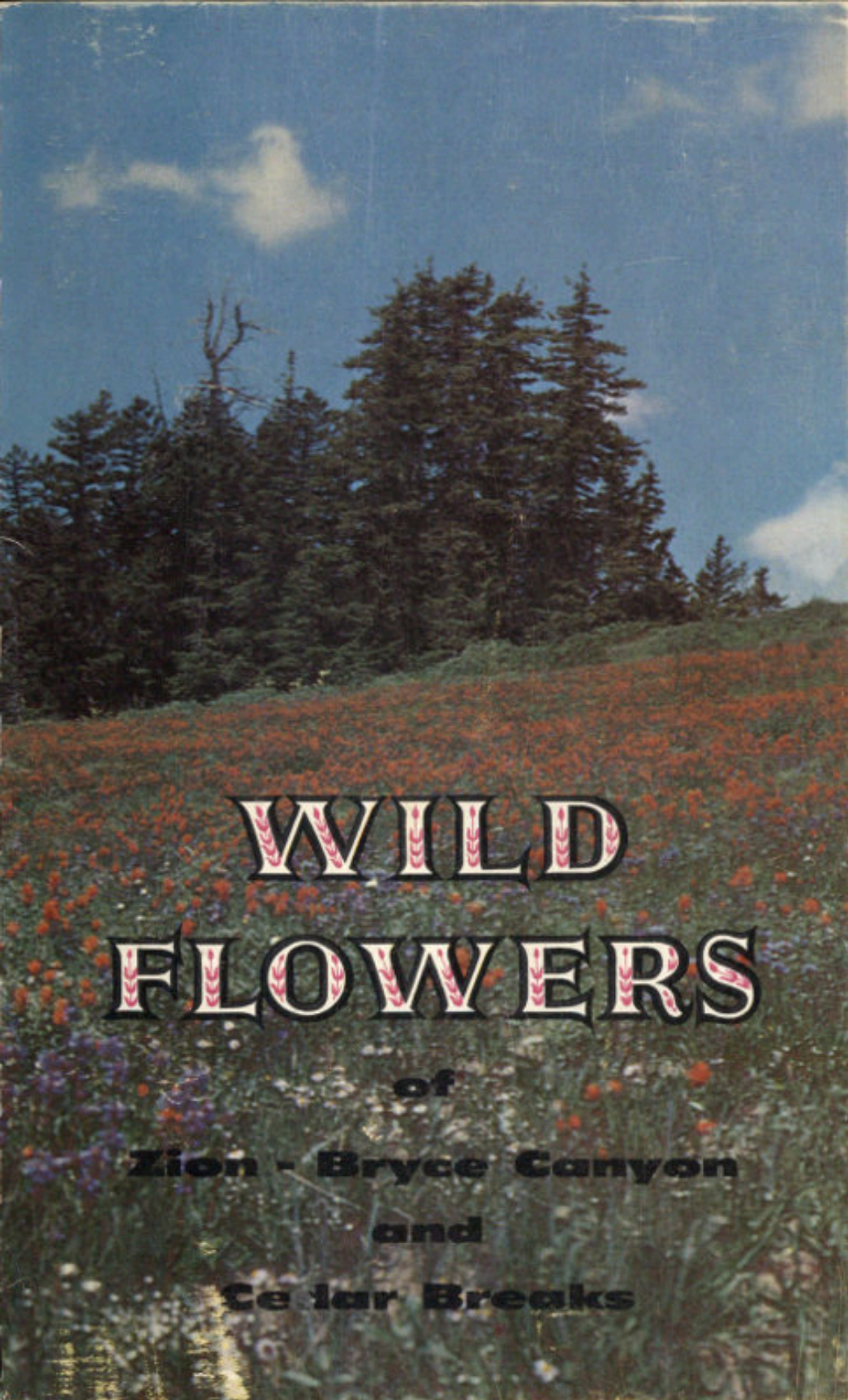
**WILD
FLOWERS**

of

Zion - Bryce Canyon

and

Cedar Breaks



WILD FLOWERS

of

Zion - Bryce Canyon

and

Cedar Breaks

The Project Gutenberg eBook of Wild Flowers of Zion and Bryce Canyon National Parks and Cedar Breaks National Monument

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

Title: Wild Flowers of Zion and Bryce Canyon National Parks and Cedar Breaks National Monument

Author: Carl E. Jepson
Leland F. Allen

Release date: August 16, 2017 [eBook #55363]
Most recently updated: October 23, 2024

Language: English

Other information and formats: www.gutenberg.org/ebooks/55363

Credits: Produced by Stephen Hutcheson and the Online Distributed Proofreading Team at <http://www.pgdp.net>

*** START OF THE PROJECT GUTENBERG EBOOK WILD
FLOWERS OF ZION AND BRYCE CANYON NATIONAL PARKS AND
CEDAR BREAKS NATIONAL MONUMENT ***

**WILD
FLOWERS
of
Zion and Bryce Canyon
NATIONAL PARKS
and
Cedar Breaks
NATIONAL MONUMENT**

Text and Color Photographs by

CARL E. JEPSON

Chief Park Naturalist

and

LELAND F. ALLEN

Park Naturalist



Published and Copyright 1958
by
ZION-BRYCE NATURAL HISTORY ASSOCIATION
in cooperation with the
NATIONAL PARK SERVICE
UTAH

INTRODUCTION



In these areas of Zion and Bryce Canyon National Parks and Cedar Breaks National Monument, four life zones are found extending from low elevations of near 3,600 feet to extremes of over 11,300 feet. The area

within elevations from the lowest point to 4,000 feet is the Lower Sonoran Zone; above that to 7,000 feet is the Upper Sonoran Zone; from 7,000 to 8,500 feet is the Transition Zone; above that to 10,000 feet is the Canadian Zone; and from 10,000 feet to the highest point on Brian Head Peak is the Hudsonian Zone.

Native plants typical of desert, mesa and mountain grow within these extremes of elevation and include a wide variety of species. Of the three areas featured, Cedar Breaks National Monument contains the greatest variety, and frequently exhibits marvelous displays, depending on the amount of rainfall from year to year.

The purpose of this booklet is to help visitors in their enjoyment of the flowers they find along the roadways and trails—flowers they observe, appreciate and probably photograph but leave UNPICKED for the pleasure of others.

Through the mediums of color photography and color lithography one hundred and six species of wild flowers and plants most commonly seen are presented in closeup detail as an aid to identification. Brief descriptions of size, habitat, blooming period, use, economic value and other details are given.

The flowers have been arranged in a general order of families, except that occasionally, for the convenience of preparing the color plates, flowers of similar color or density have been grouped on a page although they are not in the same family nor closely related. 2

The Standardized Plant Names of the American Joint Committee on Horticultural Nomenclature has been followed as the guide. In case of locally used common names, such has been indicated in the text.

This booklet on Wild Flowers has been produced by the Zion-Bryce Natural History Association in cooperation with the National Park Service. The Association has for its objective the fuller interpretation of the scenic, scientific, aesthetic and historic values of these National Parks, to the end that park visitors are provided with such information, audio and visual aids

as will help them gain a better understanding and greater appreciation of the phenomena they find in them.

It is hoped that this brief treatise on the flora may prove helpful to many visitors in learning more about some of the important species of plant life that they may discover during their visits to the Zion and Bryce Canyon National Parks and Cedar Breaks National Monument.



Credit for the photographic work goes to several individuals, all members of the interpretive staffs of these areas during the past few years. Park Naturalists Carl E. Jepson and Leland F. Allen have prepared the descriptive information given for each flower.

The printing and color processing by the Wheelwright Lithographing Company, 975 So. West Temple, Salt Lake City, Utah.

3

1. SEGOLILY MARIPOSA

Lily Family

Calochortus nuttallii

The Segolily Mariposa was chosen as the State Flower of Utah probably because of the important use early pioneers made of the root bulbs of this plant in supplementing their meager diets during the early settlement period. The name “Sego” (pronounced see-go) is of Shoshonean origin, and this flower was sacred in Indian legend long before the arrival of Utah pioneers in 1847.

This plant is found in rather dry, rocky soils and puts on a very showy display during May and June. It is more commonly known as the Mariposa Lily in other sections of the country, but in Utah it is called the Segolily.

There are three species in color of white, pink and yellow found in the Zion Region.



Segolily Mariposa

2. YELLOW MARIPOSA

Lily Family

Calochortus nuttallii var. *aureus*

Found only in the petrified forest of the Coalpits Wash section of Zion National Park, this plant is not very abundant and probably suffered from overgrazing by livestock during the settlement period before Zion became a National Park.

Mariposa in Spanish means butterfly.

This species with bright-yellow flowers is associated with a specific geologic stratum—the Petrified Forest member of the Chinle formation. It is found in great abundance in the Petrified Forest National Monument near Holbrook, Arizona.



Yellow Mariposa

3. PRAIRIE SPIDERWORT

Spiderwort Family

Tradescantia occidentalis

In the sandy areas at elevations above 4,000 feet you may find this pretty, three-petaled, deep-blue flower on its slender stalk about a foot or more in height blooming early in June. The flowers bloom at night, so are not easily found except early in the mornings. The plant is fairly abundant along the Narrows Trail, the East Rim Trail and near the East Entrance Station of Zion National Park.

Indians used the entire plant for food.



Prairie Spiderwort

5

4. BLUEDICKS

Lily Family

Dichelostemma pulchellum

This bright-blue flower, on its long, slender stalk, has a number of common names, such as Wild Hyacinth, Grass Nuts and Spanish Lily. The bulb of this plant has a nutty flavor. It was gathered by the Indians and early pioneers for food in some sections of the country. It is found sparingly in good soil areas.



Bluedicks

5. PURPLESPOT FRITILLARY

Lily Family

Fritillaria atropurpurea

A rather rare lily sometimes called Leopard Lily or Bronze Bell. Its drooping flowers on fairly tall stems are found growing in the Sagebrush areas or in alpine meadows. As they are not very conspicuous, they are often overlooked by visitors. The petals, with their mottled effect in brown,

yellow and purple spots, present a remarkable pattern of beauty when observed closely. The odor of the plant is not pleasing to humans, but is no doubt attractive to insects.



Purplespot Fritillary

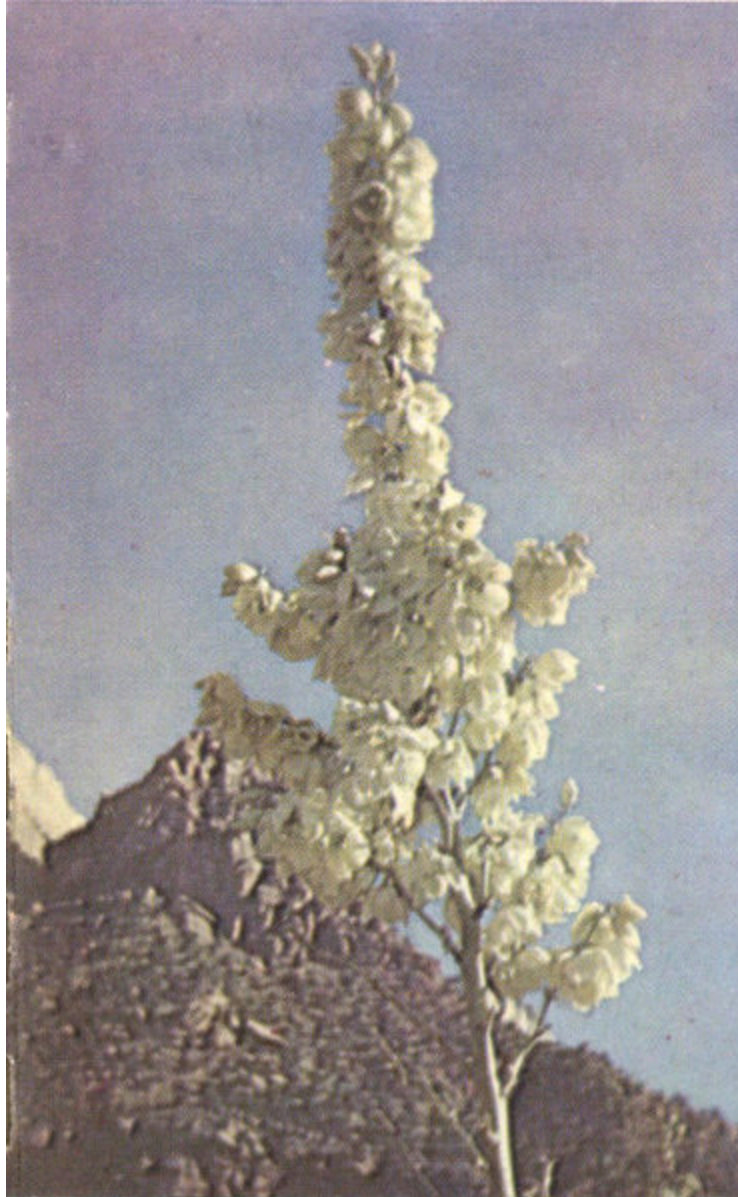
6. FINELEAF YUCCA

Yucca angustissima

Lily Family

The name “Our Lord’s Candlestick” was given to this tall, conspicuous plant of the desert by the early Spanish Padres, who were the first white men to see this region of Southern Utah. During May and June the waxy-white flowers bloom on tall stalks and soon mature into rather large seed pods.

Indians made very good use of all parts of the plant. Its fiber was used for making sandals and clothing, the seeds provided food, and the roots were used for making soap. The Navajo Indian called it Yaybi-tsa-si, which means literally “Yucca of the Gods.”



Fineleaf Yucca

7. MOUNTAIN DEATHCAMAS

Lily Family

Zigadenus elegans

An onion-like plant with a long, loose cluster of small, creamy-white flowers. Its root is shaped much like that of the onion, but is odorless. The plant is poisonous to man and beast. Deathcamas is found mostly in meadows or wet places on the plateaus where it presents a serious danger to

grazing cattle and sheep. At Cedar Breaks it blooms during July and August and is fairly common in the alpine meadows.



Mountain Deathcamas

8. WILD BUCKWHEAT

Buckwheat Family

Eriogonum umbellatum

Wild Buckwheat is commonly associated with Sagebrush and arid regions of the West. Many species of the genus are found blooming throughout the summer season.

The spreading branches grow close to the ground and help reduce erosion, and the yearly accumulation of leaves adds humus to the soil. The flower head at the top of single stalks, with its many-branched, dense cluster in a lacy pattern, makes a fine floral display of yellow. The flowers are important to the honey bee, and the ripened seeds are diligently sought by the chipmunks, other rodents and several birds.



Wild Buckwheat

9. GREEN EPHEDRA

Jointfir Family

Ephedra viridis

Not a very showy plant with its pale-green stems, very small leaves and inconspicuous flowers. It is probably of greatest interest because of the use

made of the plant by early pioneers in brewing a tea, which served as a tonic for various ailments. It was commonly called Brigham Tea, Squaw Tea or, more generally, Mormon tea.

The plant is a relative of the Pines and Firs and is very able to withstand drouth. It is found fairly abundantly in the Sagebrush and Saltbush areas of the Upper Sonoran Zone.

The drug ephedrine is obtained from some of the species of Ephedra found in China.



Green Ephedra

10. FOURWING SALTBUUSH

Goosefoot Family

Atriplex canescens

A silvery-green, profusely branched shrub growing two to five feet high with conspicuous clusters (in late summer) of four-winged seeds about the

same color as the leaves. It is very common in the Sonoran Zones and abundant in the alkaline flats of the Great Basin of Utah. In the lower portion of Zion Canyon it is abundant and often taken to be Sagebrush, which it resembles to some extent.

This plant is of value as forage for livestock, and deer feed upon it to a limited extent.



Fourwing Saltbush

9

11. CALYPSO ORCHID

Orchid Family

Calypso bulbosa

If you travel to the mountains early in June or July, you may be lucky enough to find this beautiful flower, the dainty Calypso or Fairy Slipper Orchid, as it blooms in the dense woods of Spruce and Pine or in the deep canyons along shady streams at elevations above 6,000 feet. The flowers

grow singly on stems 4 to 6 inches high and have only one large leaf. The plant takes part of its food from decaying wood or other organic matter.

Thoughtless picking of this flower has made it very rare and in danger of becoming extinct. Please do your part to help save the Fairy Slipper Orchid.



Calypso Orchid

12. FOUR-O'CLOCK

Four-O'Clock Family

Mirabilis multiflora

Closely resembling the cultivated variety of Four-O'Clock, this plant, with its abundance of brilliant magenta-colored flowers, is one of the spectacular sights in May or early June. It is a sturdy perennial with thick, glossy-green leaves spreading low over the ground. The south-facing slopes in the Sonoran Zones are its most common habitat, but it is also found in the broken lava fields. Being a night bloomer, the flowers close during the bright daylight hours and open at about four o'clock in the afternoon. Its

blooming season is generally brief, about two or three weeks, but it sometimes blooms twice in the same summer.



Four-O'Clock

13. SPRINGBEAUTY

Purslane Family

Claytonia lanceolata

Found abundantly at Cedar Breaks during May and early June and also in the shady canyons of Zion in the Transition Zone. Each plant has two narrow leaves near the base, each about 2 inches long, above which are four to five practically leafless branches with a single flower at the top of each. The plant is rarely over 6 inches high; more commonly it is flat to the ground. Flowers vary in color from white to pink or sometimes the white blossoms have pink veins or stripes which tend to accentuate their beauty. Usually one of the early blooming flowers of the high plateaus, along with the Indianpotato and Buttercup.



Springbeauty

11

14. BITTERROOT

Portulaca Family

Lewisia rediviva

This beautiful, dwarfed plant, never more than an inch or two high, is found during May mostly on the lava fields of the Transition Zone in Zion National Park. Its flower of white petals with pink veins is about 2 inches in diameter.

The Bitterroot was discovered in 1805 by the Lewis and Clark expedition while passing through western Montana. It was later named *Lewisia rediviva* by the botanist Pursh.

The plant is of economic importance to the Northwestern Indians, who discovered that the bitter, parsnip-shaped roots possess a nutritious heart of starch, which cooking reduces to a pasty mass, palatable, at least, to Indian taste.



Bitterroot

15. MARSHMARIGOLD

Buttercup Family

Caltha leptosepala

A plant of the high elevations, generally above 9,000 feet, and found abundantly at Cedar Breaks, where it comes into bloom almost as soon as the first patches of bare ground appear. Coming out usually in April or May, depending on the season, Marshmarigolds are among the earliest blooming flowers of the high elevations and often carpet the alpine meadows with a spread of white blossoms.

The white sepals, that make up the showy flower, are often mistaken for petals, which are absent. The mass of anthers of the stamens give the flower its brilliant yellow center.



Marshmarigold

16. BUTTERCUP

Buttercup Family

Ranunculus sp.

A number of species of Buttercups bloom in these areas in early April or May on the Plateaus and later in the shady canyons. In their respective areas they are often the first flowers of Spring and are followed closely by the Springbeauties or sometimes preceded by the tiny white Indianpotato of the Parsnip Family.

A thrilling sight is to find the waxy flowers of the early Buttercups at the very edge of the receding snowbanks. The blooming season for Buttercups is very brief, as a general rule, but the different species come into bloom successively.



Buttercup

13

17. COLUMBINE

Buttercup Family

Aquilegia sp.

These beautiful flowers are well known because of their wide distribution and common use as cultivated species in flower gardens. They have conspicuously shaped petals with long, hollow spurs, which contain honey and thereby attract certain insects and especially the hummingbirds. Columbines are probably the most beautiful of the native flowers of Zion, Bryce Canyon and Cedar Breaks, as they bring forth very attractive displays during the summer months. The White Columbine at Cedar Breaks is extremely large.



Columbine

18. LARKSPUR

Buttercup Family

Delphinium sp.

Larkspurs are found abundantly at Cedar Breaks during July and August in the alpine meadows of the Monument. The leaves are very similar to those of the Monkshood, but the flowers differ in color and shape. The single spur of one of the sepals is the marked feature of the Larkspur. Color and size

vary greatly for the different species; but the most common variety at Cedar Breaks is 2 to 3 feet tall, and the flowers are a purplish blue.



Larkspur

19. MONKSHOOD

Buttercup Family

Aconitum columbianum

Found abundantly at Cedar Breaks in the more open forested areas where there is partial shade and plenty of moisture. The purple Monkshood rank almost as high as their cousins the Columbines and Larkspurs in charm and beauty, with a quaintness and individuality all their own. The flower features a modified sepal shaped like a hood or helmet that protects the stamens. Extracts from the thick, turnip-shaped root are used medicinally in the treatment of certain heart diseases.



Monkshood

20. WESTERN BANEBERRY

Buttercup Family

Actaea arguta

This plant may be found in the shady forest of good moisture. It is about one to two feet tall with handsome leaves but rather inconspicuous heads of creamy-white flowers. The conspicuous bright-red or waxy-white berries about the size of peas in a dense cluster make a very attractive display in late summer. The berries are somewhat poisonous, hence the name Baneberry.



21. SAND BUTTERCUP

Buttercup Family

Ranunculus juniperinus

One of the very early blooming plants in Zion in the Transition or Upper Sonoran Zone, which often appears from mid-February to April, in bare sandy places among the Junipers and Pinyons, is the Sand Buttercup. Along the trail to the Canyon Overlook above the Great Arch is perhaps the best place to find this plant in Zion.

The flowers, a pinkish white, are found on short stems or spreading branches close to the ground. As the plants are rather small and not very showy, it takes careful searching to find them.



Sand Buttercup

22. OREGON GRAPE

Barberry Family

Berberis repens

Because of its hollylike leaves, this dwarf shrub is sometimes called the Holly Grape, but it is more commonly known as the Oregon Grape. A rather prostrate growth form accounts for a third common name; namely, Creeping Barberry.

This plant is found sparingly scattered over the areas and is probably more abundant in Zion than in Bryce Canyon or Cedar Breaks. The fruit looks very much like a cluster of grapes and is often gathered for the making of jellies or wine. The woody stems were used by the Indians in making a yellow dye.

The plants are helpful in holding the soil, as they spread close to the ground.



Oregon Grape

23. FREMONT BARBERRY

Barberry Family

Berberis fremontii

This rather tall shrub of the Sonoran Zones puts on a remarkable display in April and May with its bright yellow flowers. It is most commonly seen along the highway leading to Zion National Park from the west.

Since the plant is a secondary host of the Blackstem Rust of cereals, it is not cultivated as an ornamental shrub.

Indians used the wood of this shrub for various implements or tools. They utilized the root, which contains berberine, for a tonic, and they also made from it a brilliant-yellow dye.



Fremont Barberry

24. ELK THISTLE

Sunflower Family

Cirsium foliosum

This Thistle is widely scattered in these areas but is not very abundant. It is a stout plant, 2 to 3 feet tall, with large, prickly leaves. Its freshly budding flowers in deep pink are very attractive during the early summer. Hummingbirds and numerous insects gather food from its colorful flower head made up of many individual flowers. Some Thistles are very obnoxious and detrimental to agriculture.



Elk Thistle

25. CONEFLOWER

Sunflower Family

Rudbeckia occidentalis

At Cedar Breaks and in the high elevations of Zion and Bryce Canyon you can find this rank-growing plant in fair abundance. Its thimblelike, dark-brown flower head has numerous, inconspicuously small, yellow flowers that come out progressively up the cone from its base. The dark-brown cones, towering above the foliage of the plant, make a spectacular display against the deep-blue sky. The ripened seeds are very much sought after by rodents and numerous birds.



Coneflower

26. DESERT PRINCESPLUME

Mustard Family

Stanleya pinnata

During the months of May and June this very conspicuous plant in Zion Canyon and throughout the Sonoran Zones may be found sending up its tall spikes of lemon-yellow flowers. On the same stalk can be found the ripened and opened seed pods (siliques), fresh-blooming flowers and unopened

buds all at the same time. It has tall, stout stems, rather woody at the base, and differs from many plants in that it is tolerant of soils containing gypsum.



Desert Princesplume

27. STONECROP

Stonecrop Family

Sedum stenopetalum

Found mostly in very dry, rocky soil, these small plants, with smooth, fleshy leaves and starry-yellow flowers, are fairly conspicuous as they bloom during the early summer months. These plants have the ability to store up moisture in their fleshy leaves and stems. They are, therefore, well adapted to withstanding long periods of drouth. The plants are sometimes gathered for treatment of certain ailments.



Stonecrop

28. PRICKLYPOPPY**Poppy Family***Argemone platyceras*

You can find this plant, with its large, white flowers, most frequently along the road cuts or in abandoned fields where it has taken over as a weed. Its showy display is most abundant during midsummer or in July and August. The large, white flower, with its conspicuous yellow center, is sometimes called "The Cowboy's Fried Egg." The prickly leaves and stems account for another common name, Thistlepoppy. The plants are drought resistant and unpalatable to livestock. The seeds contain a narcotic drug more potent than opium.

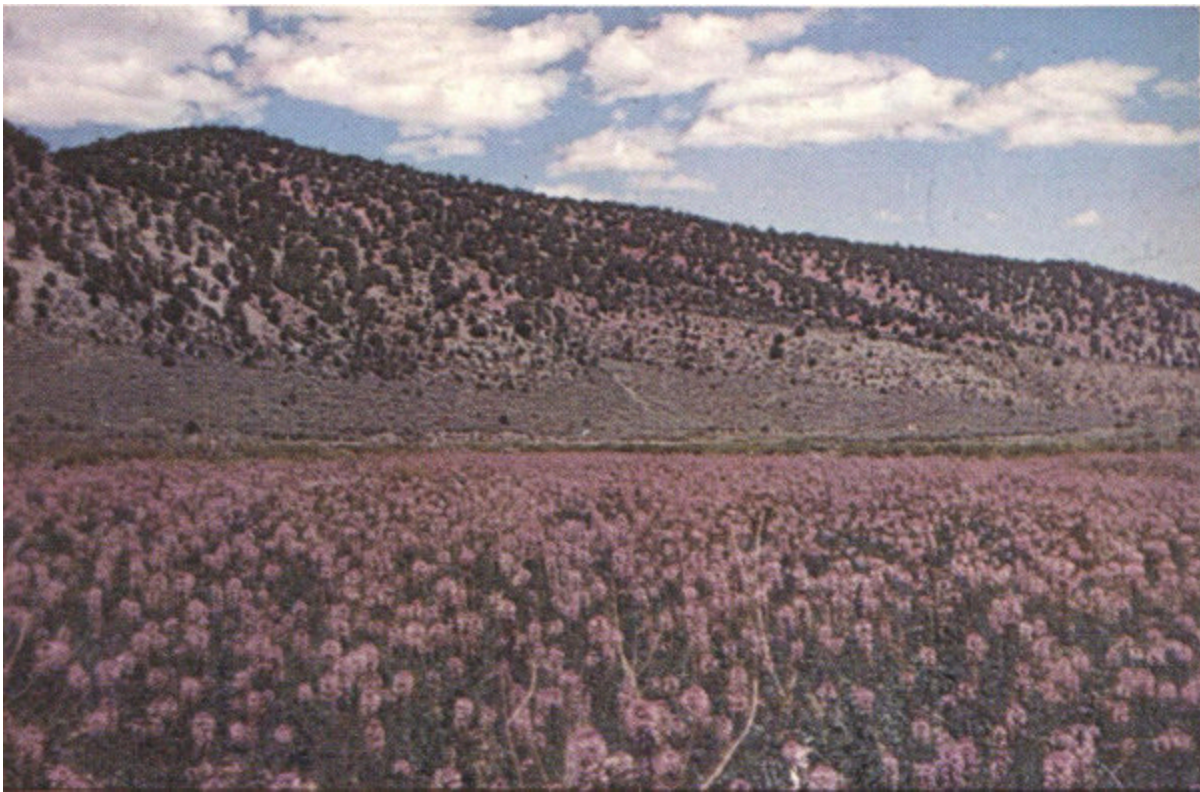


Pricklypoppy

29. ROCKY MOUNTAIN BEEPLANT**Caper Family***Cleome serrulata*

Along the roadsides of Southern Utah near these areas of Zion, Bryce Canyon and Cedar Breaks you may find this pleasing floral display of orchid or purple presented by fields of the Rocky Mountain Beeplant.

It is probably an exotic that has been brought in during recent years. Because of the unpleasant odor of crushed herbage, this plant is sometimes called skunk weed. The flowers are an important source of honey, and the seeds are eaten by a number of birds, especially the doves.



Rocky Mountain Beeplant

30. YELLOW SPIDERFLOWER

Caper Family

Cleome lutea

The plants of this genus are often called Beeplants. There are two species of Spiderflower in these areas. Yellow Spiderflower is not quite as common as the purplish-pink species commonly known as the Rocky Mountain Beeplant.

Both species are conspicuous roadside flowers in June and July. Although they are important sources of honey, they are not very sweet scented to humans. No doubt the odor helps attract insects to the flowers.



Yellow Spiderflower

21

31. ELDER

Honeysuckle Family

Sambucus racemosa

This red-berried Elder is found the most abundantly at Cedar Breaks. It is also common along the highways through forested areas of the region.

The clusters of small white flowers that come on usually in June or July give way to bright-red berries in August and present a most attractive display.

Most Elders are edible and are eaten by birds and rodents. Some people gather the berries for wines and jellies. This species, however, is considered

poisonous, and cases of poisoning have been reported from eating the berries, flowers, roots and bark.

The stalks of some Elders are pithy and fairly easily hollowed out. Indians used the stalks for making flutes.



Elder

32. LITTLELEAF MOUNTAINMAHOGANY Rose Family

Cercocarpus intricatus

This low-growing shrub is fairly important as winter browse for deer and other browsing animals. The leathery leaves are evergreen, rather narrow, pointed at both ends and curled backwards from the sides. The flowers are very small and inconspicuous, but the fruits, with their long plumes, present an interesting display. The dead wood of this shrub is very useful to campers, as it burns with extremely hot flame and gives off very little smoke.



Littleleaf Mountainmahogany

33. STANSBURY CLIFFROSE

Rose Family

Cowania stansburiana

In Zion this shrub is usually found at elevations above 4,000 feet, and it generally grows to be 6 to 8 feet high. During May and June its creamy-white flowers are suggestive of the Wild Rose. The habitat of the Cliffrose is the warm, dry slopes among the Pinyon and Juniper. Its twigs and evergreen leaves are browsed extensively by deer and other animals. The fragrance of this plant in bloom is remarkable and reminds one of Orange blossoms. The fruits are achenes and have pale, silky-haired tails 2 or more inches in length, which at times are very showy.



Stansbury Cliffrose

23

34. ANTELOPE BITTERBRUSH

Rose Family

Purshia tridentata

This shrub looks much like the Cliffrose, except for size. It is lower growing, has smaller pale-yellow flowers and its fruits are not plumed. The stubby, fan-shaped leaves are three pointed like those of the Big Sagebrush.

Bitterbrush is found most commonly on the warm, dry slopes above 4,000 feet, where it provides valuable year-round browse for deer and other animals. It is also known as Antelopebrush and Quininebush.



Antelope Bitterbrush

35. BLACKBRUSH

Rose Family

Colegyne ramosissima

A member of the Rose Family found mostly in the Sonoran Zones of Zion National Park. Several plants may be seen near the South Entrance Station. It is well named, as it has a burned and dead appearance during much of the year; however in late April and May it puts out minute grey-green leaves and creamy-yellow flowers made up of four sepals and no petals. The stamens are numerous. The Cliffrose, Bitterbrush and Mountain-mahogany are closely related to the Blackbrush.



Blackbrush

36. BUSH CINQUEFOIL

Rose Family

Potentilla fruticosa

Found most commonly at Cedar Breaks, this member of the Rose Family puts on a very showy display for a brief period of the summer, generally in July or earlier at lower elevations. It is a shrub commonly 2 to 3 feet high with leaves that are five-parted, hence the name Cinquefoil. The shrub is often dotted all over with rose-like flowers, about an inch across, with clear-yellow petals and deeper yellow anthers. The plant is common in the mountains, across the continent, up to altitudes of 10,000 feet or more.



Bush Cinquefoil

25

37. PRAIRIESMOKE

Rose Family

Geum triflorum var. ciliatum

This graceful plant, with its nodding, bell-shaped, pink-colored flowers, is found fairly abundantly in the alpine meadows of Cedar Breaks National Monument.

The plant has a number of common names such as: China Bells, Oldman-Whiskers, and Grandfather's-beard. The silvery, plumose tails of the fruit present an attractive display, especially as the sun's rays light the waving plumes in late afternoon or early morning.

Prairiesmoke plants are considered good forage for several animals, and the bumblebees gather its pollen for honey.



Prairiesmoke

38. WILD ROSE

Rose Family

Rosa sp.

There are two species of the Wild Rose in Zion and two other species in Bryce Canyon, three of which are found at Cedar Breaks National Monument.

Wild Roses are widely distributed in the northern hemisphere and are too familiar to need much description. The flowers are fairly fragrant and have bright-pink petals with a large cluster of yellow stamens. The fruit of the rose, shaped like a small apple, turns a deep-red color late in the season and adds beauty to this plant during autumn.



Wild Rose

39. SERVICEBERRY

Rose Family

Amelanchier alnifolia

One of the very early blooming shrubs or small trees in Zion Canyon, and often seen in March or April as conspicuous white masses on the canyon walls among the Pinyon and Juniper trees, is the Serviceberry.

The white flowers occur in clusters and look much like a fruit tree in bloom. The berry is shaped like a very small apple, insipid to the taste, but nevertheless used extensively by the Indians and early settlers for food. The plant has a variety of common names, including the following: Juneberry, Shadbush, Saskatoon, Sarviceberry and Pigeonberry.



Serviceberry

27

40. CHOKECHERRY

Rose Family

Prunus virginiana

Chokecherry bushes or small trees are fairly common at elevations around 6,000 feet in these areas. Early in May, and sometimes even earlier, the white and fragrant blossoms present a pleasing sight. In July or August the ripened cherries supply food for humans, many birds and small animals.

The first autumn colors are often the Chokecherry leaves as they turn scarlet, brown and red.

These shrubs are browsed heavily by cattle and some by deer, especially if other forage is scarce.



Chokecherry

41. HONEY MESQUITE

Pea Family

Prosopis juliflora

A low-growing tree of the Lower Sonoran Zone. Uncommon in Zion but fairly abundant in the desert area adjacent to the park. In early spring during March and April bright-green leaves cover the tree, and often it is laden with catkinlike clusters of greenish-yellow flowers, which attract myriads of insects including the honey bees.

The wood of this tree was used extensively by early settlers for fuel, building corrals and in making furniture and utensils. The fruit of the Mesquite, resembling a string bean, is used for food by many animals. Indians also made wide use of it by grinding the beans into a meal called "Pinole."



Honey Mesquite

42. LUPINE

Pea Family

Lupinus sp.

There are so many varieties of Lupine that it is most difficult to identify the numerous species. In this area they are found abundantly on the high plateaus, being especially plentiful at Cedar Breaks, where they fill whole meadows with a mass of blue color in midsummer.

The Lupines range in color from pale pink to deep purple, with some white, cream or yellow, but most of them are blue. Like other plants of the Pea Family, Lupines add nitrogen to the soil and thereby improve the land on which they grow. The seeds of a few species contain alkaloids which are poisonous to livestock, especially sheep.



Lupine

43. LOCO

Pea Family

Astragalus sp.

This very large genus of plants ranges from the hottest parts of the desert to high mountain peaks and far to the North. More than a dozen species are found in the Zion Region.

Some of the species contain a poisonous constituent causing, the often fatal, loco disease of livestock, particularly in horses.

Loco is a Spanish word meaning “Crazy.” Other species known as Poisonvetch, prefer soils rich in selenium, and take up enough of that toxic mineral to make them poisonous to livestock, especially sheep. The harmless species are called Milkvetch.

Nearly all the species are colorful and spectacular when in blossom, but some of them have a rank, disagreeable odor.



Loco

44. POISONVETCH

Pea Family

Astragalus sabulonum

This showy species of *Astragalus* is locally called Rattleweed because, when it is in fruit, its large, bladder-like, thin-walled pods become very brittle and give a distinct rattling sound when shaken. The pods are about one and a half inches long and heavily mottled reddish-brown in color.

The genus of *Astragalus* has been divided into three groups: Loco, Milkvetch and Poisonvetch. The species poisonous to livestock are commonly called Loco Weeds.



Poisonvetch

45. NEW MEXICO LOCUST

Pea Family

Robinia neomexicana

This shrub or small tree is fairly common in Zion Canyon. It was probably brought in by the early settlers. Its large, showy flower grows in clusters at the ends of slender branches. The blooming season is during May and June.

The tree is very thorny and has the habit of sprouting from roots or stumps and of forming dense thickets which are valuable in controlling erosion. The foliage serves as food for browsing animals, especially the deer.



New Mexico Locust

46. DESERTBEAUTY DALEA

Pea Family

Dalea johnsoni

Sometimes confused with the Desert Sage or more commonly called the Purple Sage, which it resembles to some extent, this small shrub with light-gray bark, small, gray-green leaves and terminal spikes of brilliant-purple flowers is one of the most pleasing sights in early summer.

In Zion it is found mostly in the Coalpits Wash and Shunes Creek areas and blooms generally during May. Desertbeauty is a close relative of the “Smoke Tree” of the Desert.



Desertbeauty Dalea

47. FREMONT GERANIUM

Geranium Family

Geranium fremontii

This beautiful, midsummer-blooming plant, growing about two feet high, is common on the plateaus and in the cool canyons.

The pink, veined petals, deeply lobed leaves and characteristic geranium odor help identify this plant. Some species have white flowers, but they are not common in this area. The flowers are perfect with five sepals, five petals, and five to ten stamens. The fruit is a long capsule and has given rise to the common name Cranesbill.

Cultivated Geraniums are really Pelargoniums from South Africa.



Fremont Geranium

48. ALFILERIA

Geranium Family

Erodium cicutarium

This low-growing plant, spreading close to the ground, with its finely divided leaves and small, starry-pink flowers, puts on a remarkable display in the open meadows of the large canyons. It is one of the earliest blooming species in Zion Canyon, and in seasons of abundant rain it often presents the appearance of a pale-purple lawn.

On ripening, the seed capsules split open and shoot out the seeds—each with a tiny hook in its nose and a tail with successive tight coils like a corkscrew. The seed is apparently screwed into the ground by alternating moisture and dryness which winds and unwinds the seed plume.



Alfileria

49. LEWIS FLAX

Flax Family

Linum lewisii

Here in Zion during May and June, growing along the trails on the plateaus and in cool canyons, you will find the beautiful and delicate blue flowers of the Wild Flax. The flower is nearly an inch across and has five sepals and five petals borne at the top of a slender stem having narrow leaves. At Bryce Canyon this plant is more abundant than at Zion or Cedar Breaks.

Wild Flax was named in honor of Captain Meriwether Lewis, who first discovered it, near the continental divide, during the famed Lewis and Clark expedition of 1804-1806.



Lewis Flax

50. CREOSOTEBUSH

Caltrop Family

Larrea tridentata

Probably the most characteristic shrub of the arid valleys and mesas of the Southwest is the Creosotebush, which is sometimes erroneously called Greasewood.

At times this plant fills the air with a peculiar pungent aroma that gives rise to the common name Creosotebush. Mexicans call it “Hediondilla” the (Little Bad Smeller.)

A sticky resin on the leaves is used as a poultice for bruises and sores. A resinous gum or lac deposited by scale insects on the branches is used by the Indians as a cement for fixing arrow points and mending of pottery.



Creosotebush

51. WHITEMARGIN EUPHORBIA

Spurge Family

Euphorbia albomarginata

This low-growing plant with abundant small, round or oval leaves and tiny white flowers is commonly called Carpetweed because of the manner in which it spreads over the ground.

Found mostly in the dry sandy soils, this plant serves as an excellent soil binder and is very helpful in preventing erosion.

The milky juice of this species is considered toxic and may cause dermatitis in susceptible persons. It is also known as Rattlesnake Feed and popularly supposed to be efficacious in treating snake bite.



Whitemargin Euphorbia

52. SCARLET GLOBEMALLOW

Mallow Family

Sphaeralcea coccinea

Very commonly found along the roadsides and especially prominent in the campgrounds or other disturbed areas. This member of the Mallow Family presents a lovely coral-red display as early as May, and it continues blooming throughout much of the summer.

Cotton belongs to this large and important family, which also contains such ornamental plants as the Hollyhock. Extensive fields of this flower present brilliant displays of orange.



Scarlet Globemallow

53. DESERT GLOBEMALLOW

Mallow Family

Sphaeralcea ambigua

Found most commonly along the roads and trails, this member of the Mallow Family is known by a variety of common names such as: Rose of Sharon, False Mallow and Wild Hollyhock. It generally blooms in late summer, especially in the Cedar Breaks region. The plants sometimes grow

to be 4 to 5 feet tall and have a great profusion of pink-colored flowers, really a very colorful and attractive display.



Desert Globemallow

54. FIREWEED

Evening-primrose Family

Epilobium angustifolium

This tall, willowy plant, called Fireweed, is frequently the first plant to come in after a forest fire. Its colorful bloom gives new life to the blackened ground. Fireweed is one of the world's most widely disseminated wild flowers, being found throughout much of northern North America, Europe and parts of Asia. Its seeds are scattered by the wind. In these areas it is found mostly on the high plateaus.



Fireweed

55. WOODLAND PINEDROPS

Indian Pipe Family

Pterospora andromedea

In the rich soil of the Ponderosa Pine forest of the high plateaus you may occasionally see this peculiar plant called Pinedrops. It has a single reddish-brown stalk (no green parts at all) about a foot or two high, apparently without leaves (they are mere scales), and numerous round or bell-shaped blossoms or seed pods hanging from short stems. The stalk is generally sticky with a material much like the pitch of the pine trees. It is a parasite that lives on the roots of pine trees.



Woodland Pinedrops

37

56. YELLOW VIOLET

Violet Family

Viola praemorsa

There are five different species of Violets in Zion, two of them with yellow flowers and the others with white or purple. They are found blooming in May or June on the high plateaus or sometimes later in the wet meadows of

Cedar Breaks or in shady canyons. Violets are closely related to the cultivated Pansies. The flowers are irregular, as one petal has a saclike spur which contains nectar. They have five sepals, five petals and five stamens. This species is found most commonly in the shady forest, while the purple and white species are more common in the very moist and cool areas of the shady canyons.



Yellow Violet

57. DESERT BLAZINGSTAR

Loasa Family

Mentzelia multiflora

Blazingstars, sometimes called “Stickleaf” because of their rough, hairy leaves, are plants well suited to the drought conditions of this area, as they adapt well to dry, rocky soil. They are often found in roadside cuts or other newly disturbed soils and are found blooming usually in July and August. They are conspicuous with their yellow flowers consisting of five long

petals and a large number of stamens almost as long as the petals that attract the eye as they blaze forth in the bright sunlight of midsummer. These plants are found mostly in the Transition Zone.



Desert Blazingstar

58. A HEDGEHOG CACTUS

Cactus Family

Echinocereus coccineus

This family is well represented in Zion, where ten prominent species are found, six of which are illustrated in this booklet. This species is sometimes called the Cucumber Cactus. It is found blooming in bright-red clumps as early as April or on occasion in March when the winters are not severe. Its favorite habitat appears to be the rocky slopes of the lava fields below 5,000 feet.

As cactus flowers mature into fruit, they form bulb-like bodies called tunas. The fruits of this species are about an inch or more in diameter and serve as important food for many rodents.



A Hedgehog Cactus

39

59. PURPLETORCH CACTUS

Cactus Family

Echinocereus engelmannii

Found mostly on the rocky slopes in the Lower Sonoran Zone, this cactus is fairly common and distinguished chiefly by its waxy and brilliantly colored purple flowers. The stems of this species are similar to those of the Cucumber Cactus except for being generally taller. The fruits are also similar, but the flowers differ as they are much larger and bright purple instead of red in color. They bloom generally during the month of May.



Purple Torch Cactus

60. A PRICKLY PEAR CACTUS

Cactus Family

Opuntia rhodantha

The most common cactus in Zion is this species of the Prickly Pear. It is found in a great variety of habitats, even at the higher elevations. The flowers, large and spectacular in various shades of red, salmon or yellow, bloom late in May to July. Because of its long blooming season, this species is more often found in flower than any other cactus in the park. Its pear-shaped fruit, red to purple in color, is eaten by many animals as well as by the native peoples.



A Prickly Pear Cactus

61. BEAVERTAIL CACTUS

Cactus Family

Opuntia basilaris

One species of cactus that at first sight does not seem to be a cactus is the Beavertail, for it is lacking the long spines characteristic of most cacti. Upon close examination, however, you will find that it is protected by numerous, fine spines. The name of this species is derived from its flat stems shaped somewhat like a small beaver's tail. The conspicuous magenta flowers mark this species as one of the most beautiful in the park. The fruit of this species is important as animal food, especially for chipmunks and ground squirrels.



Beavertail Cactus

62. BUCKHORN CHOLLA CACTUS

Cactus Family

Opuntia acanthocarpa

The Cholla (pronounced Choya) Cactus is the only “tree” cactus found in Zion. It is fairly common in the Lower Sonoran Zone, especially in Coalpits Wash. Strangely enough this plant is a favorite nesting place of some desert birds, including sparrows, wrens and finches. The spreading branches seem to reach out and grab the careless hiker, and the barbed spines stick so fast in the flesh that the joint of the plant is separated from the main stem before the spines can be withdrawn. Cholla blossoms come in many shades of color with the accent on yellows and reds.



Buckhorn Cholla Cactus

63. ENGLEMANN PRICKLYPEAR CACTUS Cactus Family

Opuntia engelmannii

As one of the largest of the flat-jointed or Pricklypear Cacti of Zion, this species is fairly abundant in Zion Canyon and is probably the most frequently observed species because of its dense growth in certain habitats and its fairly immense size as well as its colorful display of blossoms. It is locally called Elephant Ear Cactus. In June it produces large yellow flowers, very waxy in texture.

(See [Figure 64](#) for information concerning the fruit of this cactus.)



Engelmann Pricklypear Cactus

64. CACTUS FRUIT

Cactus Family

Opuntia engelmannii

The fruit of the Engelmann Pricklypear Cactus is similar to the fruits of many other species of cacti; it is conspicuous, being deep red in color and about the size of a large Crab Apple. It is called Tuna or locally “Cactus Apple” and ripens as early as June but more abundantly during July.

Indians of the Southwest use this fruit as an important item of their diet. Many people living in the desert have come to learn that this cactus fruit makes excellent jellies and candies. It is also feasted upon by several rodents, especially the Antelope Ground Squirrel.



Cactus Fruit

43

65. ROUNDEAF BUFFALOBERRY

Oleaster Family

Shepherdia rotundifolia

A low, evergreen shrub with small, oval leaves that appear to have been painted with a thin coat of aluminum paint through which the green shows faintly. It is most common in the Upper Sonoran Zone and may be found along the Canyon Overlook and Emerald Pool Trails in Zion National Park.

The pale-yellow flowers, not much over $\frac{1}{4}$ inch across, are often hidden by the leaves. They bloom early in April or sometimes in March. The fruit of a similar shrub has a tart berry that was gathered by the early pioneers and used as a sauce on Buffalo steaks, hence the name Buffaloberry.



Roundleaf Buffaloberry

66. WHITE EVENING-PRIMROSE

Evening-primrose
Family

Oenothera caespitosa

This member of the Evening-primrose Family, with its fragrant, large white flowers, bloom early in the evening and lasts only a few hours the next day, as it turns pink and wilts.

The flowers grow close to the ground on stemless plants. They have long, slender and hairy calyx tubes that resemble stems. This plant is found mostly in very rocky and dry habitats, which are often devoid of all other plants.



White Evening-primrose

67. YELLOW EVENING-PRIMROSE

**Evening-primrose
Family**

Oenothera strigosa

You will find this plant most commonly in the road cuts or in places where the soil has been disturbed. This species and other members of this family have very showy flowers with four broad, thin petals. Generally they bloom at night but sometimes in the daytime if growing in deep shade.

The Evening-primroses are among the comparatively few flowers blooming in Zion Canyon during the heat of midsummer, and many of them are usually found on sandy or rocky soil in the Upper Sonoran Zone.



Yellow Evening-primrose

45

68. PORTER LIGUSTICUM

Parsley Family

Ligusticum porteri

Locally called “Wild Parsley,” this fairly tall plant, with its fernlike leaves, is very common at Cedar Breaks National Monument. It grows at very high elevations. Other plants in this family, useful as food, are the carrot, parsnip, dill and anise. However, one member of the family, the Water Hemlock, is very poisonous.



Porter Ligusticum

69. GREENLEAF MANZANITA

Heath Family

Arctostaphylos patula

Many people are attracted to this plant by its bright mahogany-red bark. Its oval-shaped leaves are a bright green throughout the year. The flowers grow in clusters and sometimes are very numerous on the shrub. The fruit resembles a tiny apple, and the name Manzanita is Spanish for “Little

Apple.” Indians use the berries for food and for making a pleasant, sour drink.



Greenleaf Manzanita

70. GREEN GENTIAN

Gentian Family

Frasera speciosa

Found in the open meadows of the Cedar Breaks highlands and on the high plateaus of Zion and Bryce Canyon. The tall stalks, with their intermixture of pale-green leaves and similarly colored flowers, present a rather conspicuous sight as they tower to heights of 5 feet. The flowers have four sepals and four petals and nectar glands that attract many insects. The petals are flecked with brown and purple.



Green Gentian

71. PARRY PRIMROSE

Primrose Family

Primula parryi

This outstanding member of the Primrose Family is found only at high elevations, generally above 10,000 feet. In this region it grows fairly abundantly on Brian Head Peak but is found sparingly at Cedar Breaks. Its brilliant display of rose-red flowers is a remarkable and rewarding sight for those who gain the high places and see this alpine beauty. The smooth, thick leaves, which are quite long, all grow in a rosette at the base of the plant. The fragrance of this flower is disappointing, for it does not match its splendid color.



Parry Primrose

47

72. INDIANPOTATO

Parsnip Family

Orogenia linearifolia

As the snow melts on the plateaus, at elevations above 7,000 feet, you may find one of the very first flowers of spring. This member of the Parsnip

Family often carpets the alpine meadows with a mass of tiny white flowers never more than an inch or two high.

The root bulb was eaten by the Indians and gives rise to the common name Indianpotato. Another common name is Salt and Pepper Plant because of the speckled appearance of the flowers.

The blooming period of this plant is very brief, and soon after the flowers have faded the leaves disappear and the plant lies dormant during most of the year.



Indianpotato

73. SHOOTINGSTAR

Primrose Family

Dodecatheon pauciflorum

Shootingstars are one of the early blooming flowers in the alpine meadows of Cedar Breaks and on the high plateaus. They also come on very early in the moist canyons of Zion. Along with the Columbine and Monkeyflower they are the predominant plants of the Hanging Gardens found on many of the Canyon Walls.

The basal leaves spread close to the ground, while the flowers in a variety of colors, white, pink or purple grow on stems 6 to 8 inches high. The down-pointed stamens of the flower center and the reflexed or turned-back petals gives the flower its common name Shootingstar.



Shootingstar

74. FRINGED GENTIAN

Gentian Family

Gentiana thermalis

One of the most beautiful of all mountain flowers, the Fringed Gentian is commonly found in the moist meadows of Cedar Breaks at elevations near or above 10,500 feet. The flower stalks are generally 6 to 10 inches tall, and each bears a handsome flower about two inches long with four fringed petals. At times the Fringed Gentian carpets the alpine meadows with a waving mass of deep-blue color. This species is the Park Flower of Yellowstone National Park.



Fringed Gentian

75. BUTTERFLY MILKWEED

Milkweed Family

Asclepias tuberosa

There are four fairly common species of Milkweed in Zion, but the one pictured is the most common. This variety is found in the dry places above 4,000 feet and is especially abundant in Birch Creek Canyon.

The conspicuous orange flowers grow on fairly tall stalks about two feet in height and make this plant very easy to find. The stems are quite hairy, leafy and contain a milky juice. As the fruits develop in large boat-shaped pods, the seeds burst forth bearing long, silky hairs that assist the wind in scattering them over wide areas.



Butterfly Milkweed

76. PINK PHLOX

Phlox Family

Phlox canescens

Four kinds of phlox are common in Zion, and other species are found at Cedar Breaks and Bryce Canyon. The plants are generally small, but the flowers are fairly showy with their five bright, pink-colored petals. They are found generally in dry, rocky soils and bloom mostly in early summer or during May and June. Sometimes Phlox will cover extensive areas with a bright carpet of pink or white. A special display of Phlox may be found on

the summit of Brian Head Peak near Cedar Breaks. These plants are very helpful in holding the soil and in preventing erosion.



Pink Phlox

77. SKYROCKET GILIA

Phlox Family

Gilia aggregata

This plant is found most commonly in the Ponderosa Pine belt of these areas where its star-shaped, scarlet flower adds a bit of brilliance to the scene. The individual flowers, with their long, tubular corollas and star-shaped petals, are masterpieces of beauty. Their shape and color have given rise to such common names as Trumpet Phlox and Scarlet Gilia.

In their search for nectar, hummingbirds are noticeably attracted to the flowers of the plant. Birds and insects in taking the nectar, help in the pollination of many flowers.



Skyrocket Gilia

78. MOUNTAIN BLUEBELLS

Borage Family

Mertensia ciliata

A fairly tall perennial growing to be from 2 to 3 feet high and found only in moist places at high elevations of the plateaus. It is common at and near Cedar Breaks along the streams or in the swampy meadows. The small clusters of drooping, delicate-blue flowers come mostly in May and June or even later at the higher elevations of Brian Head Peak near Cedar Breaks. Before maturing the flowers may be pinkish to white in color.



Mountain Bluebells

79. NARROWLEAF PUCCOON

Borage Family

Lithospermum incisum

This member of the Borage Family is well adapted to the dry habitats. The plants are commonly found in clumps, but they are generally widely scattered rather than in dense growths as in the case of the Bluebells or Mertensias that belong to the same Family. Its showy trumpetlike yellow flowers attract many insects as they bloom during April and May. The seeds are hard, white and shiny, hence the name *Lithospermum*, meaning stone seed.



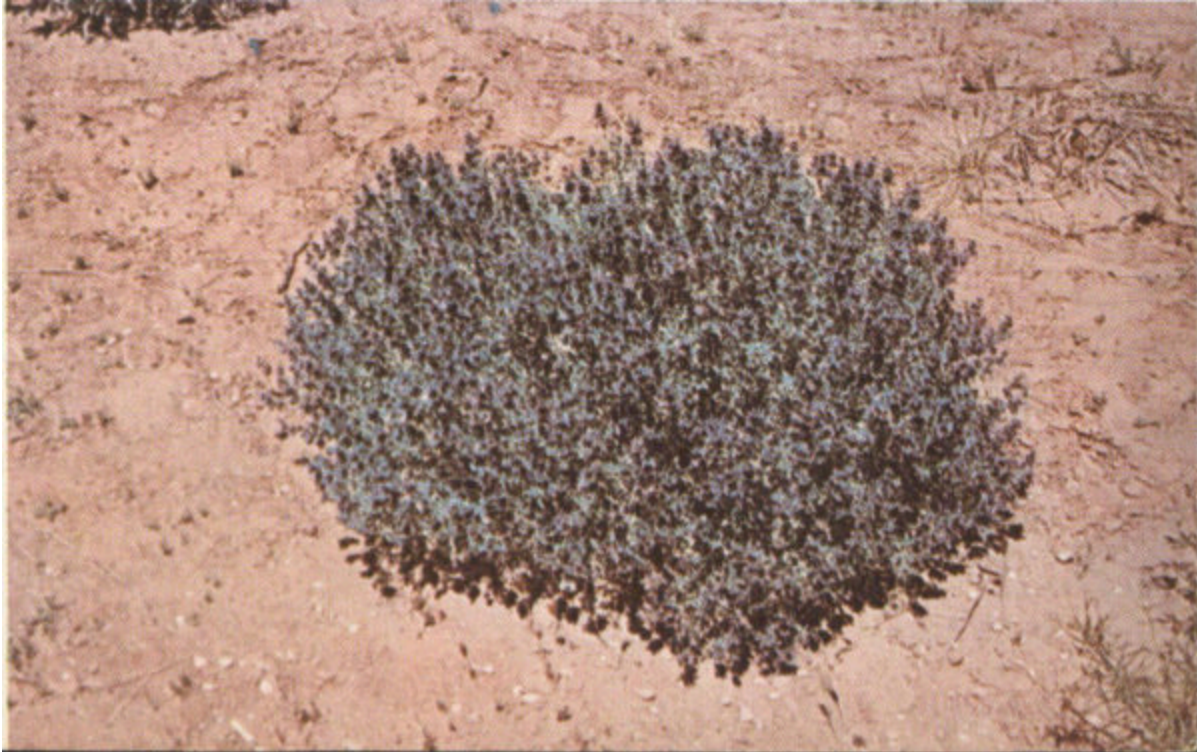
Narrowleaf Puccoon

80. DESERT SAGE

Mint Family

Salvia carnosa

Although looking very much like a clump of Sagebrush, this small shrub is a member of the Mint Family and not closely related to the Common Sagebrush. The clusters of bright purple flowers, as they come on in May or June, certainly add brilliant adornment to this sage-green plant. It is fairly common in the Sonoran Zones and well scattered throughout much of the Southwest. This is the plant referred to in the storied Land of the Purple Sage. One of the best places to look for this plant is along the Emerald Pool Trail in Zion National Park.



Desert Sage

81. MULLEIN

Figwort Family

Verbascum thapsus

This rather unusual-appearing plant is not a native but rather an introduced species from Europe that has spread over most of Western America. It is fairly common along the roads and trails.

The Hopi Indians are said to dry and smoke the leaves, and this is thought to cure people who are mentally unbalanced.

Early Greeks and Romans dipped dried mullein stalks in tallow to make lampwicks. The English named it “Torchweed” and the Spanish called it “Candelaria.”



Mullein

82. ROYAL PENSTEMON

Figwort Family

Penstemon speciosus

This species, also known as Purple Penstemon, is found in much the same habitats as the Blue Penstemon. It is generally abundant along roadsides at elevations above 4,500 feet. It is fairly common at Cedar Breaks and at

Bryce Canyon. The flowers vary in color from deep blue to dark purple, and the stalks vary in height from 12 to 20 inches.



Royal Penstemon

83. EATON PENSTEMON

Penstemon eatoni

Figwort Family

Found mostly in the cool canyons, this plant is sometimes confused with the Skyrocket Gilia or the Western Cardinalflower, which it resembles somewhat. Penstemon usually has a greater number of blooms on each flower stalk than do these flowers that appear like it. This variety is not nearly as common as many other Penstemons in these areas. It is far more brilliantly colored, however, which accounts for such common names as Scarlet Bugler and Scarlet Penstemon.



Eaton Penstemon

84. THICKLEAF PENSTEMON

Figwort Family

Penstemon pachyphyllus

The penstemons are sometimes called Wild Snapdragons because of the close resemblance to related cultivated species. They are also called Beardstongue because one of the five stamens is covered with numerous hairs. This blue variety, Thickleaf Penstemon, is found mostly at higher elevations or on the plateaus where they bloom during June and July.



Thickleaf Penstemon

85. PALMER PENSTEMON**Figwort Family***Penstemon palmeri*

This is one of the very beautiful and conspicuous flowers of Zion National Park. The flowers are borne on tall spikes and are brightly colored. This is the largest and most common Penstemon found along the trails and roadways of Zion below 6,000 feet. It is especially common in freshly disturbed soils as the road cuts. The leaves are grey-green, and each pair with the exception of the lower ones is joined at the base, creating the impression that it is one leaf with the stem growing through the center. The fairly large flowers, which are pale lavender in color, begin blooming in May and are found in bloom throughout most of the summer.



Palmer Penstemon

86. SACRED DATURA**Potato Family**

Datura meteloides

A conspicuous plant with very large, white, sometimes lavender-tinted flowers that open at night and wilt in the bright morning sunlight. A single plant may have as many as 100 blooms at one time. Datura is one of the few plants that blooms during the hot summer in Zion Canyon. Many different names are locally applied to it, and include the following: Zion Lily, Moon Lily, Jimson Weed and Thornapple. The plant is poisonous to eat, especially the seeds, and was used by several Indian tribes to induce stupor and dreams as a part of a widespread religious cult. It contains a deadly narcotic principle akin to atropine.



Sacred Datura

87. MONKEYFLOWER

Mimulus cardinalis

Figwort Family

One of the very beautiful flowering plants along the Zion Narrows Trail and in cool, damp places of the shady canyons is this Crimson Monkeyflower with its orange-red blossoms and deep-green leaves. Its flowers are 1 to 2 inches long and the wide-toothed leaves are 3 to 5 inches long. It is the largest Monkeyflower in the park. Some plants are found blooming throughout much of the summer season, especially along the canyon walls where there are seeps of water most of the year.



Monkeyflower

57

88. INDIAN PAINTBRUSH

Figwort Family

Castilleja coccinea

From early March until May the warm hillsides below 6,000 feet are made brilliant by clumps of deep-red flowers often found next to patches of Mountain Mahogany. They are the Dwarf Indian Paintbrush and are by far the most conspicuous early spring flower in Zion. Found very abundantly

along the park road from the East Entrance to the Zion Tunnel, they present a most pleasing sight early in the season. Other species are very plentiful at Cedar Breaks and often carpet the meadows in showy orange or red. Several species are also found at Bryce Canyon National Park.



Indian Paintbrush

89. SNOWBERRY

Honeysuckle Family

Symphoricarpos utahensis

This low, spreading shrub is recognized by its shreddy bark, small, oval, opposite leaves on very short petioles, and in late summer or fall by its white berries. The small, pinkish flowers are not at all conspicuous and are often overlooked. The plant is browsed by deer and other animals and is sometimes called Buckbrush. It is found mostly in the high plateaus of Zion and is fairly common at Cedar Breaks and Bryce Canyon. The fruit, although very showy, is not very tasty.



Snowberry

90. BEARBERRY HONEYSUCKLE Honeysuckle Family
Lonicera involucrata

This member of the Honeysuckle Family has a number of common names such as Twinberry Honeysuckle, Ink-berry and Pigeon-bush.

The flowers, which are yellow and always come in pairs, are very attractive to the hummingbirds. Mature fruits are black berries about the size of peas and are partially enclosed by reddish bracts. The plants are unpalatable and browsed very slightly, but the fruits are eaten by birds and chipmunks.



Bearberry Honeysuckle

59

91. ELEPHANTHEAD PEDICULARIS

Figwort Family

Pedicularis groenlandica

You will find this strange-looking plant in the wet meadows of the alpine areas of Cedar Breaks and on the Plateaus of the Kolob Section and the Horse Pasture Plateau of Zion National Park. Its blooming season is July and August. The peculiarly modified petals of the corolla resemble the forehead, ears and waving trunk of an elephant, hence the common name Elephanthead.



Elephanthead Pedicularis

92. AMERICAN HAREBELL

Bellflower Family

Campanula rotundifolia

In the drier habitats of the alpine regions around Cedar Breaks and generally along the roadsides in large clumps you may find this beautiful blue flower. The lovely deep-blue flowers, drooping on their hairlike stems, have such perfect shape and simple grace. The Harebell is very wide spread,

being found in Scotland, Northern Europe and Asia as well as over much of North America.



American Harebell

93. WESTERN CARDINALFLOWER

Lobelia splendens

Lobelia Family

A pleasant surprise to many park visitors is to find this spectacular flower, with its abundant scarlet blooms on long stalks, presenting a colorful display during the late summer when most plants have ceased blooming. This colorful species, known also as Scarlet Lobelia, is very abundant along the Narrows Trail of Zion Canyon and also along water courses of other shady canyons. The long, tubular corollas and pointed petals arranged in irregular pattern of two and three identify this plant from the Scarlet Penstemon which it resembles.



Western Cardinalflower

94. RABBITBRUSH

Sunflower Family

Chrysothamnus sp.

Rabbitbrush is fairly common along roadways and trails in the arid regions. Because rabbits find this plant a favorite shelter, it has been named Rabbitbrush. Indians boil the plant for yellow dye, and white man found certain species of *Chrysothamnus* to contain rubber.

Consideration was given to the production of rubber from Rabbitbrush during the First World War and up until the discovery of synthetic rubber.



Rabbitbrush

61

95. CURLYCUP GUMWEED

Sunflower Family

Grindelia squarrosa

An exotic plant probably brought into the area by vehicles, as it is found most commonly along the roadways or in cultivated fields. Once started it spreads very rapidly along the highways or in cultivated areas.

The plant is suspected to be toxic to livestock, but is rarely eaten. It is used in the treatment of asthma in humans. In addition, external use is made of it to relieve the irritation caused by Poison Ivy.

The plants have probably migrated to this area from the central plains states.



Curlycup Gumweed

96. PURPLE ASTER

Sunflower Family

Aster sp.

The Asters and Fleabanes are sometimes confused, but they can generally be recognized by the difference in the number of ray flowers. That is, Asters have only about half as many ray flowers as do the Fleabanes.

Species of Purple Aster form an important part of the late summer floral display at Cedar Breaks and Bryce Canyon. They come on after the Lupine, Columbine, Indian Paintbrush and other early bloomers have faded.



Purple Aster

97. FLEABANE

Sunflower Family

Erigeron speciosus

There are numerous species of Fleabane in this region. Some particular kind may be found at any time of the growing season, for certain species bloom very early and others continue late in autumn.

Some species of Fleabane grow in dense masses and, in early spring, carpet the meadows and roadsides. The ray flowers of the Fleabanes are generally twice as numerous per head as are the ray flowers of the Asters. The plants are quite similar in other respects.



Fleabane

63

98. MOUNTAIN SUNFLOWER

Sunflower Family

Helianthella uniflora

Sunflowers abound in these parks during the late summer. As the early summer flowers, mostly in blues and purples, fade, the yellow and red flowers come on. This is especially true in the alpine meadows of Cedar Breaks in August when the predominant species are the Sunflowers.

The seeds of the Sunflowers supply abundant food for many birds and small mammals.



Mountain Sunflower

99. COMMON SUNFLOWER

Sunflower Family

Helianthus annuus

The very large flowers of these plants sometimes present a colorful display as they take over the roadways or abandoned fields. Members of this group are generally considered weeds because of their habit of crowding out the more desirable species. Certain species of Sunflowers are now being developed for commercial purposes and produce oil for cooking and meal for livestock feed. Indians in some areas of North America cultivated sunflowers for food and for trade.



Common Sunflower

100. GAILLARDIA

Sunflower Family

Gaillardia parryi

This handsome and conspicuous plant is found growing in the Sonoran Zones of Zion. It has a slender, rough stalk, about a foot tall, and stiff, rather hairy, dull-green leaves growing mostly from the root. The beautiful flowers, about three inches across, have golden-yellow rays which are three pointed. The center of the flower is a shaded maroon and yellow, very velvety, and becoming an attractive fuzzy, round, purplish head when the rays drop off. This plant blooms mostly in May and June.



Gaillardia

65

101. DESERT MARIGOLD

Sunflower Family

Baileya multiradiata

These golden-yellow flowers, measuring about three inches across, are fairly common along the trails and roadways of Zion Canyon and in other low-elevation areas of the park. They bloom during May and June. The ray flowers become bleached and papery as they mature, thus accounting for the name Paper Daisy. This attractive composite is also known as Desert Baileya. In California this plant is cultivated for the flower trade. It has been found poisonous to sheep, although horses crop the flower heads, apparently without harmful effects.



Desert Marigold

102. WESTERN YARROW

Sunflower Family

Achillea lanulosa

Western Yarrow is very widespread and can be found in much of Western America. It is more common at elevations above 5,000 feet in these areas. It grows to be 12 to 20 inches high, and the fernlike leaves, which have a pungent odor when crushed, and the umbrella-shaped clusters of flowers are characteristics of this plant that help to identify it.

Since ancient times the plant has been highly regarded for its healing properties. Legend ascribed the discovery of this virtue to Achilles, in whose honor the plant is named.



Western Yarrow

103. HEARTLEAF ARNICA

Sunflower Family

Arnica cordifolia

A common flower in the Pine and Spruce forest of Cedar Breaks and the alpine areas of Zion and Bryce Canyon is the Heartleaf Arnica. The yellow ray flowers are few, while the disk or central flowers of the flower head are numerous. The flowers measure about three inches across and are often mistaken for sunflowers. The heart-shaped leaves help distinguish this flower from its close relatives.

Tincture of arnica is obtained from certain species of Arnica.



Heartleaf Arnica

67

104. MEADOW SALSIFY

Sunflower Family

Tragopogon pratensis

This interesting plant, known also as Oyster Plant, has been naturalized from Europe and is now quite common in the West. It has a smooth, stout hollow stem about 2 feet tall, rather dark-green, smooth leaves clasping at the base, and handsome flowers from 2 to 4 inches across. The flowers open early in the morning, closing at midday, to remain closed until the next morning.

Meadow Salsify is most commonly found along the roadways or in other places where the native soil has been disturbed.

See [Figure 105](#) for description of the seed of this flower.



Meadow Salsify

105. MEADOW SALSIFY

Sunflower Family

Tragopogon pratensis

This habitat view of the Meadow Salsify illustrates one of the important ways in which plants scatter their seeds about. As the flower matures into seeds in a conspicuous and very large, dandelion-like head, each seed is equipped with a perfect parachute of silky fibers. Winds often sweep these flight-equipped seeds for many miles and result in wide dissemination of this species, which was introduced into this country from Europe not very long ago.

The seeds of many plants are scattered about in various ways—some by wind, others by water and many by the birds and animals.



Meadow Salsify Fruit

106. ARROWLEAF BALSAMROOT Sunflower Family

Balsamorhiza sagittata

This plant, with its large, showy yellow flowers, is often found on the southern exposures of steep hillsides or in the Sagebrush flats. It was first discovered by Lewis and Clark on their expedition across the continent in 1804-1806.

The rind of the root contains a turpentiney balsam, but the heart of the root is edible and was used by the Indians and early pioneers. The plant is called Mormon Biscuit in Utah. The seeds of the plant were used by the Indians to make "Pinole" or meal, and the stems and leaves were eaten as greens.



Arrowleaf Balsamroot

69

107. WESTERN WALLFLOWER

Mustard Family

Erysimum capitatum

There are two kinds of Wallflowers in Zion National Park. Their bright-yellow flowers, which grow on stalks taller than those of most other mustards, make them among the most attractive members of this family. They are usually found on rather dry slopes in the Upper Sonoran and Transition Zones.

Notice how the petals are arranged as a cross which is a characteristic of all members of the Cruciferae or Mustard Family.



Western Wallflower

108. BITTERCRESS

Mustard Family

Cardamine hirsuta

You may find this plant blooming during April and May in the Sonoran Zones of Zion National Park. Its habitat is generally the dry sandy hillsides rather than the deep canyons.

The wide-spreading, circular, doom-shaped clumps present an attractive display in pure white flowers. The petals of four are arranged like a cross.

Being a perennial, the clumps seem to expand from year to year and often reach a spread of four to five feet across. The plants are useful in building soil and in preventing erosion.



Bittercress

109. HUMMINGBIRD TRUMPET Evening-primrose Family

Zauschneria garrettii

One of the late blooming plants in Zion National Park is the Hummingbird Trumpet, also called Fire-chalice, or sometimes the Wild Fuchsia. It can often be found on the Canyon Overlook Trail or on the West Rim Trail at elevations near 6,000 feet.

It can be identified by the narrow oval leaves pointed and toothed, and the fushialike flowers, narrowly funnel-shaped, with the pistil and stamens extending beyond the petals.

The brilliant scarlet of this flower in fairly dense clusters makes a very attractive display in late August and September.



Hummingbird Trumpet

INDEX

Common Name	Scientific Name	Figure Number
A		
Alfilera	<i>Erodium cicutarium</i>	<u>48</u>
American Harebell	<i>Campanula rotundifolia</i>	<u>92</u>
Antelope Bitterbrush	<i>Purshia tridentata</i>	<u>34</u>
Arnica, Heartleaf	<i>Arnica cordifolia</i>	<u>103</u>
Arrowleaf Balsamroot	<i>Balsamorhiza sagittata</i>	<u>106</u>
Aster, Purple	<i>Aster sp.</i>	<u>96</u>
B		
Balsamroot, Arrowleaf	<i>Balsamorhiza sagittata</i>	<u>106</u>
Baneberry, Western	<i>Actaea arguta</i>	<u>20</u>
Bearberry Honeysuckle	<i>Lonicera involucrata</i>	<u>90</u>
Beavertail Cactus	<i>Opuntia basilaris</i>	<u>61</u>
Bitterbrush, Antelope	<i>Purshia tridentata</i>	<u>34</u>
Bittercress	<i>Cardamine hirsuta</i>	<u>108</u>
Bitterroot	<i>Lewisia rediviva</i>	<u>14</u>
Blackbrush	<i>Coleogyne ramosissima</i>	<u>35</u>
Blazingstar, Desert	<i>Mentzelia multiflora</i>	<u>57</u>
Bluebells, Mountain	<i>Mertensia ciliata</i>	<u>78</u>
Bluedicks	<i>Dichelostemma pulchellum</i>	<u>4</u>
Buckhorn Cholla Cactus	<i>Opuntia acanthocarpa</i>	<u>62</u>
Buckwheat, Wild	<i>Eriogonum umbellatum</i>	<u>8</u>
Buffaloberry, Roundleaf	<i>Shepherdia rotundifolia</i>	<u>65</u>
Bush Cinquefoil	<i>Potentilla fruticosa</i>	<u>36</u>
Buttercup, Sand	<i>Ranunculus juniperinus</i>	<u>21</u>
Buttercup	<i>Ranunculus sp.</i>	<u>16</u>

Butterfly Milkweed	<i>Asclepias tuberosa</i>	75
--------------------	---------------------------	--------------------

C

Cactus Fruit	<i>Opuntia engelmannii</i>	64
--------------	----------------------------	--------------------

Calypso Orchid	<i>Calypso bulbosa</i>	11
----------------	------------------------	--------------------

Cardinalflower, Western	<i>Lobelia splendens</i>	93
-------------------------	--------------------------	--------------------

Cinquefoil, Bush	<i>Potentilla fruticosa</i>	36
------------------	-----------------------------	--------------------

Chokecherry	<i>Prunus virginiana</i>	40
-------------	--------------------------	--------------------

Cliffrose, Stansbury	<i>Cowania stansburiana</i>	33
----------------------	-----------------------------	--------------------

Columbine	<i>Aquilegia sp.</i>	17
-----------	----------------------	--------------------

Coneflower	<i>Rudbeckia occidentalis</i>	25
------------	-------------------------------	--------------------

Creosotebush	<i>Larrea tridentata</i>	50
--------------	--------------------------	--------------------

Curlycup Gumweed	<i>Grindelia squarrosa</i>	95
------------------	----------------------------	--------------------

D

Deathcamas, Mountain	<i>Zigadenus elegans</i>	7
----------------------	--------------------------	-------------------

Desertbeauty Dalea	<i>Dalea johnsoni</i>	46
--------------------	-----------------------	--------------------

Desert Blazingstar	<i>Mentzelia multiflora</i>	57
--------------------	-----------------------------	--------------------

Desert Globemallow	<i>Sphaeralcea ambigua</i>	53
--------------------	----------------------------	--------------------

Desert Marigold	<i>Baileya multiradiata</i>	101
-----------------	-----------------------------	---------------------

Desert Princessplume	<i>Stanleya pinnata</i>	26
----------------------	-------------------------	--------------------

Desert Sage	<i>Salvia carnosa</i>	80
-------------	-----------------------	--------------------

E

Eaton Penstemon	<i>Penstemon eatoni</i>	83
-----------------	-------------------------	--------------------

Elder, Red-berried	<i>Sambucus racemosa</i>	31
--------------------	--------------------------	--------------------

Elephanthead Pedicularis	<i>Pedicularis groenlandica</i>	91
--------------------------	---------------------------------	--------------------

Elk Thistle	<i>Cirsium foliosum</i>	24
-------------	-------------------------	--------------------

Engelmann Pricklypear	<i>Opuntia engelmannii</i>	64
-----------------------	----------------------------	--------------------

Cactus

Ephedra, Green	<i>Ephedra viridis</i>	9
----------------	------------------------	-------------------

Euphorbia, Whitemargin	<i>Euphorbia albomarginata</i>	51
------------------------	--------------------------------	--------------------

Evening-primrose, White	<i>Oenothera caespitosa</i>	66
-------------------------	-----------------------------	--------------------

Evening-primrose, Yellow	<i>Oenothera strigosa</i>	67
--------------------------	---------------------------	--------------------

F

Fineleaf Yucca	<i>Yucca angustissima</i>	<u>6</u>
Firechalice	<i>Zauschneria garrettii</i>	<u>109</u>
Fireweed	<i>Epilobium angustifolium</i>	<u>54</u>
Flax, Lewis	<i>Linum lewisii</i>	<u>49</u>
Fleabane	<i>Erigeron speciosus</i>	<u>97</u>
Four-O'Clock	<i>Mirabilis multiflora</i>	<u>12</u>
Fourwing Saltbush	<i>Atriplex canescens</i>	<u>10</u>
Fremont Geranium	<i>Geranium fremontii</i>	<u>47</u>
Fremont Barberry	<i>Berberis fremonti</i>	<u>23</u>
Fringed Gentian	<i>Gentiana thermalis</i>	<u>74</u>
Fritillary, Purplespot	<i>Fritillaria atropurpurea</i>	<u>5</u>

G

Gaillardia	<i>Gaillardia parryi</i>	<u>100</u>
Gentian, Fringed	<i>Gentiana thermalis</i>	<u>74</u>
Gentian, Green	<i>Frasera speciosa</i>	<u>70</u>
Geranium, Fremont	<i>Geranium fremontii</i>	<u>47</u>
Gilia, Skyrocket	<i>Gilia aggregata</i>	<u>77</u>
Globemallow, Scarlet	<i>Sphaeralcea coccinea</i>	<u>52</u>
Globemallow, Desert	<i>Sphaeralcea ambigua</i>	<u>53</u>
Green Ephedra	<i>Ephedra viridis</i>	<u>9</u>
Green Gentian	<i>Frasera speciosa</i>	<u>70</u>
Greenleaf Manzanita	<i>Arctostaphylis patula</i>	<u>69</u>
Gumweed, Curlyleaf	<i>Grindelia squarrosa</i>	<u>95</u>

H

Harebell, American	<i>Campanula petiolata</i>	<u>92</u>
Heartleaf Arnica	<i>Arnica cordifolia</i>	<u>103</u>
Hedgehog Cactus	<i>Echinocereus coccineus</i>	<u>58</u>
Honey Mesquite	<i>Prosopis juliflora</i>	<u>41</u>
Honeysuckle, Bearberry	<i>Lonicera involucrata</i>	<u>90</u>
Hummingbird Trumpet	<i>Zauschneria garrettii</i>	<u>109</u>

I

Indianpotato	<i>Orogenia linearifolia</i>	<u>72</u>
Indian Paintbrush	<i>Castilleja coccinea</i>	<u>88</u>
L		
Larkspur	<i>Delphinium sp.</i>	<u>18</u>
Lewis Flax	<i>Linum lewisii</i>	<u>49</u>
Ligusticum, Porter	<i>Ligusticum porteri</i>	<u>68</u>
Littleleaf	<i>Cercocarpus intricatus</i>	<u>32</u>
Mountainmahogany		
Loco	<i>Astragalus sp.</i>	<u>43</u>
Locust, New Mexico	<i>Robinia neomexicana</i>	<u>45</u>
Lupine	<i>Lupinus sp.</i>	<u>42</u>
M		
Manzanita, Greenleaf	<i>Arctostaphylis patula</i>	<u>69</u>
Mariposa, Segolily	<i>Calochortus nuttallii</i>	<u>1</u>
Mariposa, Yellow	<i>Calochortus nuttallii</i> var. <i>aureus</i>	<u>2</u>
Marigold, Desert	<i>Baileya multiradiata</i>	<u>101</u>
Marshmarigold	<i>Caltha leptosepala</i>	<u>15</u>
Meadow Salsify	<i>Tragopogon pratensis</i>	<u>104-105</u>
Mesquite, Honey	<i>Prosopis juliflora</i>	<u>41</u>
Milkweed, Butterfly	<i>Asclepias tuberosa</i>	<u>75</u>
Monkeyflower	<i>Mimulus cardinalis</i>	<u>87</u>
Monkshood	<i>Aconitum columbianum</i>	<u>19</u>
Mountain Bluebells	<i>Mertensia ciliata</i>	<u>78</u>
Mountain Deathcamas	<i>Zigadenus elegans</i>	<u>7</u>
Mountainmahogany, Littleleaf	<i>Cercocarpus intricatus</i>	<u>32</u>
Mountain Sunflower	<i>Helianthella uniflora</i>	<u>98</u>
Mullein	<i>Verbascum thapsus</i>	<u>81</u>
N		
New Mexico Locust	<i>Robinia neomexicana</i>	<u>45</u>

O

Oregon Grape	<i>Berberis repens</i>	<u>22</u>
Oysterplant - Meadow	<i>Tragopogon pratensis</i>	<u>104-105</u>
Salsify		

P

Palmer Penstemon	<i>Penstemon palmeri</i>	<u>85</u>
Parry Primrose	<i>Primula parryi</i>	<u>71</u>
Penstemon, Eaton	<i>Penstemon eatoni</i>	<u>83</u>
Penstemon, Palmer	<i>Penstemon palmeri</i>	<u>85</u>
Penstemon, Royal	<i>Penstemon speciosus</i>	<u>82</u>
Penstemon, Thickleaf	<i>Penstemon pachyphyllus</i>	<u>84</u>
Phlox, Pink	<i>Phlox canescens</i>	<u>76</u>
Pinedrops, Woodland	<i>Pterospora andromedea</i>	<u>55</u>
Poisonvetch	<i>Astragalus sabulosus</i>	<u>44</u>
Porter Ligusticum	<i>Ligusticum porteri</i>	<u>68</u>
Prairiesmoke	<i>Geum triflorum</i> var. <i>ciliatum</i>	<u>37</u>
Prairie Spiderwort	<i>Tradescantia occidentalis</i>	<u>3</u>
Pricklepoppy	<i>Argemone platyceras</i>	<u>28</u>
Pricklypear Cactus	<i>Opuntia rhodantha</i>	<u>60</u>
Princesplume, Desert	<i>Stanleya pinnata</i>	<u>26</u>
Puccoon, Narrowleaf	<i>Lithospermum incisum</i>	<u>79</u>
Purplespot Fritillary	<i>Fritillaria atropurpurea</i>	<u>5</u>
Purple Torch Cactus	<i>Echinocereus engelmannii</i>	<u>59</u>

R

Rabbitbrush	<i>Chrysothamnus</i> sp.	<u>94</u>
Rocky Mountain Beeplant	<i>Cleome serrulata</i>	<u>29</u>
Rose, Wild	<i>Rosa</i> sp.	<u>38</u>
Roundleaf Buffaloberry	<i>Shepherdia rotundifolia</i>	<u>65</u>
Royal Penstemon	<i>Penstemon speciosus</i>	<u>82</u>

S

Sacred Datura	<i>Datura meteloides</i>	<u>86</u>
Sage, Desert	<i>Salvia carnosa</i>	<u>80</u>
Saltbush, Fourwing	<i>Atriplex canescens</i>	<u>10</u>

Sand Buttercup	<i>Ranunculus juniperinus</i>	<u>21</u>
Scarlet Globemallow	<i>Sphaeralcea coccinea</i>	<u>52</u>
Segolily Mariposa	<i>Calochortus nuttalli</i>	<u>1</u>
Serviceberry	<i>Amelanchier alnifolia</i>	<u>39</u>
Shootingstar	<i>Dodecatheon pauciflorum</i>	<u>73</u>
Skyrocket Gilia	<i>Gilia aggregata</i>	<u>77</u>
Spiderflower, Yellow	<i>Cleome lutea</i>	<u>30</u>
Snowberry	<i>Symphoricarpos utahensis</i>	<u>89</u>
Spiderwort, Prairie	<i>Tradescantia occidentalis</i>	<u>3</u>
Springbeauty	<i>Claytonia lanceolata</i>	<u>13</u>
Stansbury Cliffrose	<i>Cowania stansburiana</i>	<u>33</u>
Stonecrop	<i>Sedum stenopetalum</i>	<u>27</u>
Sunflower, Common	<i>Helianthus annuus</i>	<u>99</u>
Sunflower, Mountain	<i>Helianthella uniflora</i>	<u>98</u>

T

Thickleaf Penstemon	<i>Penstemon pachyphyllus</i>	<u>84</u>
Thistle, Elk	<i>Cirsium foliosum</i>	<u>107</u>

V

Violet, Yellow	<i>Viola praemorsa</i>	<u>56</u>
----------------	------------------------	---------------------------

W

Western Baneberry	<i>Actaea arguta</i>	<u>20</u>
Western Cardinalflower	<i>Lobelia splendens</i>	<u>93</u>
Western Wallflower	<i>Erysimum capitatum</i>	<u>107</u>
Western Yarrow	<i>Achillea lanulosa</i>	<u>102</u>
Whitemargin Euphorbia	<i>Euphorbia albomarginata</i>	<u>51</u>
Wild Buckwheat	<i>Eriogonum umbellatum</i>	<u>8</u>
Wild Rose	<i>Rosa sp.</i>	<u>38</u>
Woodland Pinedrops	<i>Pterospora andromedea</i>	<u>55</u>

Y

Yarrow, Western	<i>Achillea lanulosa</i>	<u>102</u>
Yellow Mariposa	<i>Calochortus nuttallii</i> var. <i>aureus</i>	<u>2</u>

Yellow Spiderflower	<i>Cleome lutea</i>	30
Yellow Violet	<i>Viola praemorsa</i>	56
Yucca, Fineleaf	<i>Yucca angustissima</i>	6

REFERENCES

Arnberger, Leslie P. (Drawings by Jeanne R. Janish)
Flowers of the Southwest Mountains. Southwestern Monuments Assoc.,
Globe, Arizona, 1952.

Armstrong, Margaret
Field Book of Western Wild Flowers. G. P. Putnam's Sons, 1915.

Bailey, H. E. and Bailey, V. L.
Woody Plants of the Western National Parks. The University Press, Notre
Dame, Indiana, 1949.

Benson, Lyman
The Cacti of Arizona. University of New Mexico Press, 1950.

Clements, Frederic E. and Clements, Edith S.
Rocky Mountain Flowers. The H. W. Wilson Co., New York, 1928.

Coulter, John M. and Nelson, Aven
New Manual of Botany of the Central Rocky Mountains. American Book
Company, New York, 1919.

Dodge, Natt N. (Drawings by Jeanne R. Janish)
Flowers of the Southwest Deserts. Southwestern Monuments Association,
Globe, Arizona, 1954.

Jaeger, Edmund C.
Desert Wild Flowers. Stanford University Press, Revised Edition, 1944.

Joint Committee on Horticultural Nomenclature
Standardized Plant Names. Second Edition 1942. J. Horace McFarland Co.

Kearney, Thomas H. and Pebble, Robert H.

Arizona Flora. University of California Press, 1951.

McDougal, W. B. and Baggley, Herma A.
Plants of Yellowstone National Park. Revised Edition, Yellowstone Library
and Museum Association, 1956.

McDougal, W. B. and Sperry, Omar E.
Plants of Big Bend National Park. Government Printing Office, 1951.

McDougal, W. B.
Checklist of Plants of Grand Canyon National Park. Bulletin No. 10. Grand
Canyon Natural History Association, 1947.

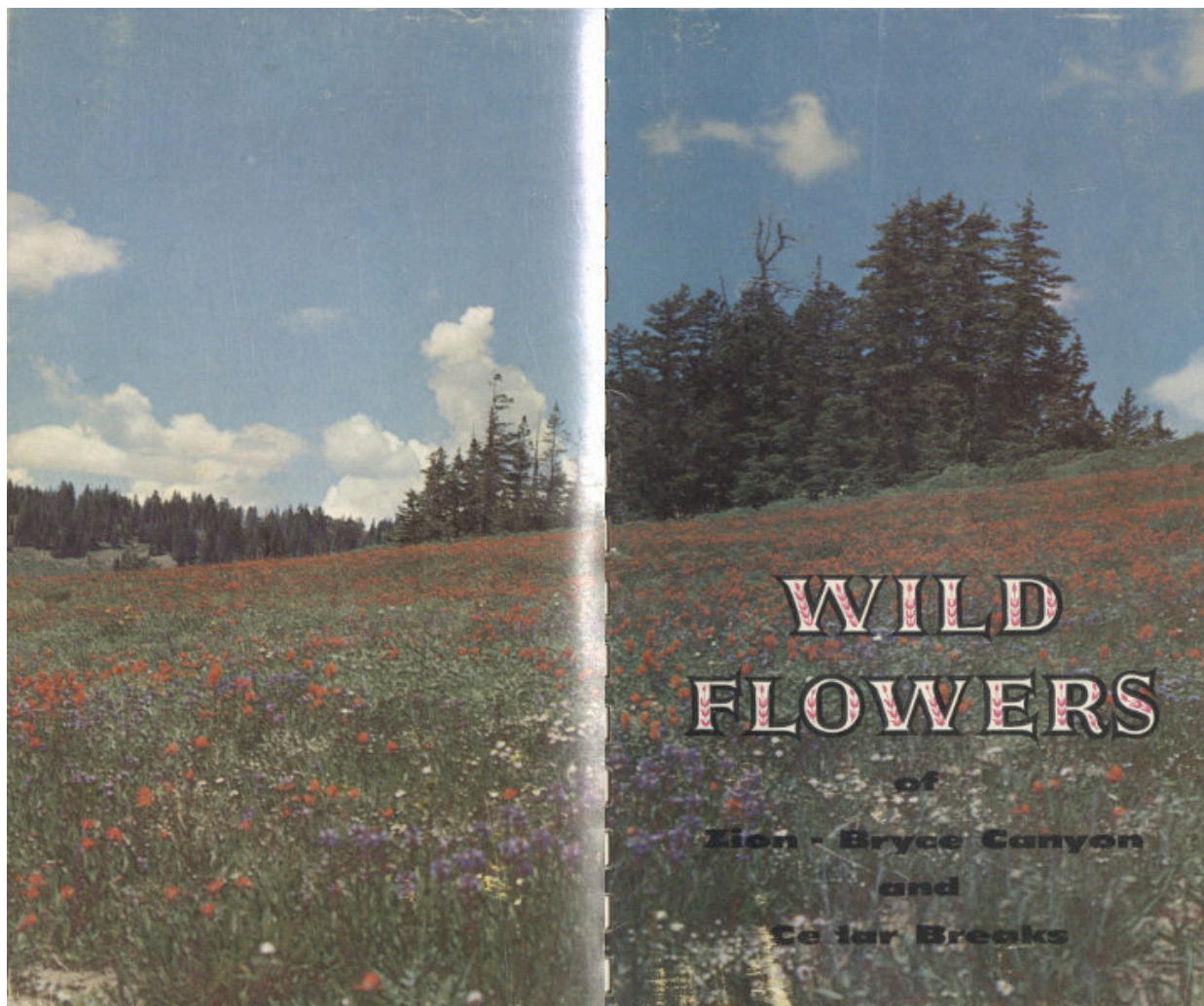
Patraw, Pauline M. (Drawings by Jeanne R. Janish)
Flowers of the Southwest Mesas. Southwestern Monuments Association,
Globe, Arizona, 1954.

Presnall, C. C. and Patraw, Pauline M.
Plants of Zion National Park. Bulletin No. 1. Zion-Bryce Natural History
Association, 1937.

Preston, Richard J.
Rocky Mountain Trees. The Iowa State College Press, Ames, Iowa, 1947.

Saunders, Charles Francis
Western Wild Flowers and Their Stories. Doubleday, Doran Co., 1933.

Tidestrom, Ivar
Flora of Utah and Nevada. Government Printing Office, 1925.



Transcriber's Notes

- Silently corrected a few typos.
- Retained publication information from the printed edition: this eBook is public-domain in the country of publication.
- In the text versions only, text in italics is delimited by _underscores_.

*** END OF THE PROJECT GUTENBERG EBOOK WILD FLOWERS
OF ZION AND BRYCE CANYON NATIONAL PARKS AND CEDAR
BREAKS NATIONAL MONUMENT ***

Updated editions will replace the previous one—the old editions will be renamed.

Creating the works from print editions not protected by U.S. copyright law means that no one owns a United States copyright in these works, so the Foundation (and you!) can copy and distribute it in the United States without permission and without paying copyright royalties. Special rules, set forth in the General Terms of Use part of this license, apply to copying and distributing Project Gutenberg™ electronic works to protect the PROJECT GUTENBERG™ concept and trademark. Project Gutenberg is a registered trademark, and may not be used if you charge for an eBook, except by following the terms of the trademark license, including paying royalties for use of the Project Gutenberg trademark. If you do not charge anything for copies of this eBook, complying with the trademark license is very easy. You may use this eBook for nearly any purpose such as creation of derivative works, reports, performances and research. Project Gutenberg eBooks may be modified and printed and given away—you may do practically ANYTHING in the United States with eBooks not protected by U.S. copyright law. Redistribution is subject to the trademark license, especially commercial redistribution.

START: FULL LICENSE

THE FULL PROJECT GUTENBERG™ LICENSE

PLEASE READ THIS BEFORE YOU DISTRIBUTE OR USE THIS WORK

To protect the Project Gutenberg™ mission of promoting the free distribution of electronic works, by using or distributing this work (or any other work associated in any way with the phrase “Project Gutenberg”), you agree to comply with all the terms of the Full Project Gutenberg License available with this file or online at www.gutenberg.org/license.

Section 1. General Terms of Use and Redistributing Project Gutenberg electronic works

1.A. By reading or using any part of this Project Gutenberg electronic work, you indicate that you have read, understand, agree to and accept all the terms of this license and intellectual property (trademark/copyright) agreement. If you do not agree to abide by all the terms of this agreement, you must cease using and return or destroy all copies of Project Gutenberg electronic works in your possession. If you paid a fee for obtaining a copy of or access to a Project Gutenberg electronic work and you do not agree to be bound by the terms of this agreement, you may obtain a refund from the person or entity to whom you paid the fee as set forth in paragraph 1.E.8.

1.B. “Project Gutenberg” is a registered trademark. It may only be used on or associated in any way with an electronic work by people who agree to be bound by the terms of this agreement. There are a few things that you can do with most Project Gutenberg electronic works even without complying with the full terms of this agreement. See paragraph 1.C below. There are a lot of things you can do with Project Gutenberg electronic works if you follow the terms of this agreement and help preserve free future access to Project Gutenberg electronic works. See paragraph 1.E below.

1.C. The Project Gutenberg Literary Archive Foundation (“the Foundation” or PGLAF), owns a compilation copyright in the collection of Project Gutenberg electronic works. Nearly all the individual works in the collection are in the public domain in the United States. If an individual work is unprotected by copyright law in the United States and you are

located in the United States, we do not claim a right to prevent you from copying, distributing, performing, displaying or creating derivative works based on the work as long as all references to Project Gutenberg are removed. Of course, we hope that you will support the Project Gutenberg mission of promoting free access to electronic works by freely sharing Project Gutenberg works in compliance with the terms of this agreement for keeping the Project Gutenberg name associated with the work. You can easily comply with the terms of this agreement by keeping this work in the same format with its attached full Project Gutenberg License when you share it without charge with others.

1.D. The copyright laws of the place where you are located also govern what you can do with this work. Copyright laws in most countries are in a constant state of change. If you are outside the United States, check the laws of your country in addition to the terms of this agreement before downloading, copying, displaying, performing, distributing or creating derivative works based on this work or any other Project Gutenberg work. The Foundation makes no representations concerning the copyright status of any work in any country other than the United States.

1.E. Unless you have removed all references to Project Gutenberg:

1.E.1. The following sentence, with active links to, or other immediate access to, the full Project Gutenberg License must appear prominently whenever any copy of a Project Gutenberg work (any work on which the phrase “Project Gutenberg” appears, or with which the phrase “Project Gutenberg” is associated) is accessed, displayed, performed, viewed, copied or distributed:

This eBook is for the use of anyone anywhere in the United States and most other parts of the world at no cost and with almost no restrictions whatsoever. You may copy it, give it away or re-use it under the terms of the Project Gutenberg™ License included with this eBook or online at www.gutenberg.org. If you are not located in the United States, you will have to check the laws of the country where you are located before using this eBook.

1.E.2. If an individual Project Gutenberg electronic work is derived from texts not protected by U.S. copyright law (does not contain a notice indicating that it is posted with permission of the copyright holder), the work can be copied and distributed to anyone in the United States without paying any fees or charges. If you are redistributing or providing access to a work with the phrase “Project Gutenberg” associated with or appearing on the work, you must comply either with the requirements of paragraphs 1.E.1 through 1.E.7 or obtain permission for the use of the work and the Project Gutenberg trademark as set forth in paragraphs 1.E.8 or 1.E.9.

1.E.3. If an individual Project Gutenberg electronic work is posted with the permission of the copyright holder, your use and distribution must comply with both paragraphs 1.E.1 through 1.E.7 and any additional terms imposed by the copyright holder. Additional terms will be linked to the Project Gutenberg License for all works posted with the permission of the copyright holder found at the beginning of this work.

1.E.4. Do not unlink or detach or remove the full Project Gutenberg License terms from this work, or any files containing a part of this work or any other work associated with Project Gutenberg.

1.E.5. Do not copy, display, perform, distribute or redistribute this electronic work, or any part of this electronic work, without prominently displaying the sentence set forth in paragraph 1.E.1 with active links or immediate access to the full terms of the Project Gutenberg License.

1.E.6. You may convert to and distribute this work in any binary, compressed, marked up, nonproprietary or proprietary form, including any word processing or hypertext form. However, if you provide access to or distribute copies of a Project Gutenberg work in a format other than “Plain Vanilla ASCII” or other format used in the official version posted on the official Project Gutenberg website (www.gutenberg.org), you must, at no additional cost, fee or expense to the user, provide a copy, a means of exporting a copy, or a means of obtaining a copy upon request, of the work in its original “Plain Vanilla ASCII” or other form. Any alternate format must include the full Project Gutenberg License as specified in paragraph 1.E.1.

1.E.7. Do not charge a fee for access to, viewing, displaying, performing, copying or distributing any Project Gutenberg works unless you comply with paragraph 1.E.8 or 1.E.9.

1.E.8. You may charge a reasonable fee for copies of or providing access to or distributing Project Gutenberg electronic works provided that:

- You pay a royalty fee of 20% of the gross profits you derive from the use of Project Gutenberg works calculated using the method you already use to calculate your applicable taxes. The fee is owed to the owner of the Project Gutenberg trademark, but he has agreed to donate royalties under this paragraph to the Project Gutenberg Literary Archive Foundation. Royalty payments must be paid within 60 days following each date on which you prepare (or are legally required to prepare) your periodic tax returns. Royalty payments should be clearly marked as such and sent to the Project Gutenberg Literary Archive Foundation at the address specified in Section 4, “Information about donations to the Project Gutenberg Literary Archive Foundation.”
- You provide a full refund of any money paid by a user who notifies you in writing (or by e-mail) within 30 days of receipt that s/he does not agree to the terms of the full Project Gutenberg™ License. You must require such a user to return or destroy all copies of the works possessed in a physical medium and discontinue all use of and all access to other copies of Project Gutenberg™ works.
- You provide, in accordance with paragraph 1.F.3, a full refund of any money paid for a work or a replacement copy, if a defect in the electronic work is discovered and reported to you within 90 days of receipt of the work.
- You comply with all other terms of this agreement for free distribution of Project Gutenberg™ works.

1.E.9. If you wish to charge a fee or distribute a Project Gutenberg™ electronic work or group of works on different terms than are set forth in this agreement, you must obtain permission in writing from the Project Gutenberg Literary Archive Foundation, the manager of the Project

Gutenberg™ trademark. Contact the Foundation as set forth in Section 3 below.

1.F.

1.F.1. Project Gutenberg volunteers and employees expend considerable effort to identify, do copyright research on, transcribe and proofread works not protected by U.S. copyright law in creating the Project Gutenberg™ collection. Despite these efforts, Project Gutenberg™ electronic works, and the medium on which they may be stored, may contain “Defects,” such as, but not limited to, incomplete, inaccurate or corrupt data, transcription errors, a copyright or other intellectual property infringement, a defective or damaged disk or other medium, a computer virus, or computer codes that damage or cannot be read by your equipment.

1.F.2. LIMITED WARRANTY, DISCLAIMER OF DAMAGES - Except for the “Right of Replacement or Refund” described in paragraph 1.F.3, the Project Gutenberg Literary Archive Foundation, the owner of the Project Gutenberg™ trademark, and any other party distributing a Project Gutenberg™ electronic work under this agreement, disclaim all liability to you for damages, costs and expenses, including legal fees. YOU AGREE THAT YOU HAVE NO REMEDIES FOR NEGLIGENCE, STRICT LIABILITY, BREACH OF WARRANTY OR BREACH OF CONTRACT EXCEPT THOSE PROVIDED IN PARAGRAPH 1.F.3. YOU AGREE THAT THE FOUNDATION, THE TRADEMARK OWNER, AND ANY DISTRIBUTOR UNDER THIS AGREEMENT WILL NOT BE LIABLE TO YOU FOR ACTUAL, DIRECT, INDIRECT, CONSEQUENTIAL, PUNITIVE OR INCIDENTAL DAMAGES EVEN IF YOU GIVE NOTICE OF THE POSSIBILITY OF SUCH DAMAGE.

1.F.3. LIMITED RIGHT OF REPLACEMENT OR REFUND - If you discover a defect in this electronic work within 90 days of receiving it, you can receive a refund of the money (if any) you paid for it by sending a written explanation to the person you received the work from. If you received the work on a physical medium, you must return the medium with your written explanation. The person or entity that provided you with the defective work may elect to provide a replacement copy in lieu of a refund. If you received the work electronically, the person or entity providing it to

you may choose to give you a second opportunity to receive the work electronically in lieu of a refund. If the second copy is also defective, you may demand a refund in writing without further opportunities to fix the problem.

1.F.4. Except for the limited right of replacement or refund set forth in paragraph 1.F.3, this work is provided to you 'AS-IS', WITH NO OTHER WARRANTIES OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO WARRANTIES OF MERCHANTABILITY OR FITNESS FOR ANY PURPOSE.

1.F.5. Some states do not allow disclaimers of certain implied warranties or the exclusion or limitation of certain types of damages. If any disclaimer or limitation set forth in this agreement violates the law of the state applicable to this agreement, the agreement shall be interpreted to make the maximum disclaimer or limitation permitted by the applicable state law. The invalidity or unenforceability of any provision of this agreement shall not void the remaining provisions.

1.F.6. INDEMNITY - You agree to indemnify and hold the Foundation, the trademark owner, any agent or employee of the Foundation, anyone providing copies of Project Gutenberg™ electronic works in accordance with this agreement, and any volunteers associated with the production, promotion and distribution of Project Gutenberg™ electronic works, harmless from all liability, costs and expenses, including legal fees, that arise directly or indirectly from any of the following which you do or cause to occur: (a) distribution of this or any Project Gutenberg work, (b) alteration, modification, or additions or deletions to any Project Gutenberg work, and (c) any Defect you cause.

Section 2. Information about the Mission of Project Gutenberg

Project Gutenberg is synonymous with the free distribution of electronic works in formats readable by the widest variety of computers including obsolete, old, middle-aged and new computers. It exists because of the

efforts of hundreds of volunteers and donations from people in all walks of life.

Volunteers and financial support to provide volunteers with the assistance they need are critical to reaching Project Gutenberg's goals and ensuring that the Project Gutenberg collection will remain freely available for generations to come. In 2001, the Project Gutenberg Literary Archive Foundation was created to provide a secure and permanent future for Project Gutenberg and future generations. To learn more about the Project Gutenberg Literary Archive Foundation and how your efforts and donations can help, see Sections 3 and 4 and the Foundation information page at www.gutenberg.org.

Section 3. Information about the Project Gutenberg Literary Archive Foundation

The Project Gutenberg Literary Archive Foundation is a non-profit 501(c)(3) educational corporation organized under the laws of the state of Mississippi and granted tax exempt status by the Internal Revenue Service. The Foundation's EIN or federal tax identification number is 64-6221541. Contributions to the Project Gutenberg Literary Archive Foundation are tax deductible to the full extent permitted by U.S. federal laws and your state's laws.

The Foundation's business office is located at 41 Watchung Plaza #516, Montclair NJ 07042, USA, +1 (862) 621-9288. Email contact links and up to date contact information can be found at the Foundation's website and official page at www.gutenberg.org/contact

Section 4. Information about Donations to the Project Gutenberg Literary Archive Foundation

Project Gutenberg™ depends upon and cannot survive without widespread public support and donations to carry out its mission of increasing the number of public domain and licensed works that can be freely distributed in machine-readable form accessible by the widest array of equipment

including outdated equipment. Many small donations (\$1 to \$5,000) are particularly important to maintaining tax exempt status with the IRS.

The Foundation is committed to complying with the laws regulating charities and charitable donations in all 50 states of the United States. Compliance requirements are not uniform and it takes a considerable effort, much paperwork and many fees to meet and keep up with these requirements. We do not solicit donations in locations where we have not received written confirmation of compliance. To SEND DONATIONS or determine the status of compliance for any particular state visit www.gutenberg.org/donate.

While we cannot and do not solicit contributions from states where we have not met the solicitation requirements, we know of no prohibition against accepting unsolicited donations from donors in such states who approach us with offers to donate.

International donations are gratefully accepted, but we cannot make any statements concerning tax treatment of donations received from outside the United States. U.S. laws alone swamp our small staff.

Please check the Project Gutenberg web pages for current donation methods and addresses. Donations are accepted in a number of other ways including checks, online payments and credit card donations. To donate, please visit: www.gutenberg.org/donate.

Section 5. General Information About Project Gutenberg electronic works

Professor Michael S. Hart was the originator of the Project Gutenberg concept of a library of electronic works that could be freely shared with anyone. For forty years, he produced and distributed Project Gutenberg eBooks with only a loose network of volunteer support.

Project Gutenberg eBooks are often created from several printed editions, all of which are confirmed as not protected by copyright in the U.S. unless a

copyright notice is included. Thus, we do not necessarily keep eBooks in compliance with any particular paper edition.

Most people start at our website which has the main PG search facility:
www.gutenberg.org.

This website includes information about Project Gutenberg, including how to make donations to the Project Gutenberg Literary Archive Foundation, how to help produce our new eBooks, and how to subscribe to our email newsletter to hear about new eBooks.